Conducting Unconventional Monetary Policy with Foreign Exchange Reserves*

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Abstract

This paper studies sterilized asset purchase programs in emerging markets and developing economies. Sterilized asset purchases are unconventional monetary policy tools implemented for the first time during the recent COVID-19 crisis. The paper provides a theoretical framework to examine the effectiveness and design of this new policy tool. The model economy is vulnerable to sudden stops due to financial market imperfection and liability dollarization. In a sudden stop, the balance sheet effect is triggered, causing large contractions in real activity. Instead of constrained domestic banks, the government plays a key role in funding intermediation. Asset purchases accompanied by sales of FX reserves relax banks' leverage constraints, breaking down the balance sheet effect. The policy effectively mitigates the impact of a sudden stop, improving welfare. The trade-offs are also discussed. Deep contractions in real activity can be avoided with large-scale asset purchases. They may, however, impede the economy's recovery. In terms of policy design, purchasing corporate bonds sterilized with foreign exchange reserves is the most effective among the types of sterilized asset purchases considered.

JEL Classification: E52, E58, F32, F41

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1 Introduction

Emerging market and developing economies (EMDEs) are vulnerable to sharp reversals in capital inflows, known as *sudden stops* (Dornbusch and Werner 1994; Calvo 1998). In a sudden stop, the country risk premium rises and the exchange rate depreciates, generating severe financial market disruptions and large contractions in real economic activity (Krugman 1999; Aghion, Bacchetta and Banerjee 2004; Céspedes, Chang and Velasco 2004). Policy responses to sudden stops often draw on unconventional monetary policy tools aimed at providing liquidity to financial markets (Chang and Velasco 2017).

More recently, during the sudden stop episode of the COVID-19 crisis, EMDEs implemented asset purchase programs for the first time. One important policy consideration was to implement this unconventional tool in a way that it complements the conventional interest rate policy. In particular, if asset purchases are not sterilized, market rates may deviate from the policy rate, potentially weakening the commitment to the inflation-targeting mandate.[1]

The design of sterilized asset purchase programs, however, remains an open question: which assets to purchase, how large the purchases should be, and how to sterilize them. This paper addresses these questions by developing a theoretical framework.

Empirical motivation Sterilization was a central concern for EMDE policymakers during the COVID-19 crisis. Governor Lesetja Kganyago of the South African Reserve Bank (SARB) emphasized this concern in his lecture (Kganyago 2020). He noted that persistent and large-scale unsterilized asset purchases increase bank reserves and are "likely to be inflationary" in EMDEs. He also stressed that, without sterilization, the overnight rate would deviate from the policy rate. Operations are therefore required "to keep the actual overnight rate close to the repo target." The SARB implemented sterilized asset purchases during the crisis: it purchased government bonds in the secondary market amounting to R25 billion (approximately 0.6 percent of GDP), sterilizing the resulting increase in reserves through its deposit rate facility.

Figure 1 illustrates the actual use of asset purchase programs across EMDEs in the early period of the COVID-19 crisis. The vertical axis reports purchases as a percentage of 2020 GDP conducted from March to August 2020. The blue bars indicate purchases of government bonds in primary and secondary markets, while the orange bars indicate purchases of private sector assets, including corporate bonds, asset-backed securities, and exchange-traded funds. The policy details varied across countries in terms of the size and composition of purchases and the means of sterilization. For example, the Philippines conducted the largest government bond purchases, amounting to 5.7 percent of GDP. Chile purchased only private assets (2.1 percent of GDP), while Colombia and Hungary purchased both government and private assets (1.1 and 0.9 percent of GDP, respectively). Sterilization was implemented through different instruments, including sales of foreign assets (e.g., Croatia), security issuance (e.g., Poland), and deposit facilities (e.g., South Africa).[2]

[1] The 1997 Asian Financial Crisis has taught EMDEs a key lesson about the importance of sterilization. In the crisis, unsterilized liquidity provision created inflation expectations, intensifying the effects of the crisis (Burnside, Eichenbaum and Rebelo 2001). The idea of sterilization of asset purchases is not new. See Swanson (2011) and Kuttner (2018) for the details of the Operational Twist implemented for the first time by the Federal Reserve in 1961.

[2] See International Monetary Fund (2020); Adrian et al. (2021); Arena et al. (2021) for the details.

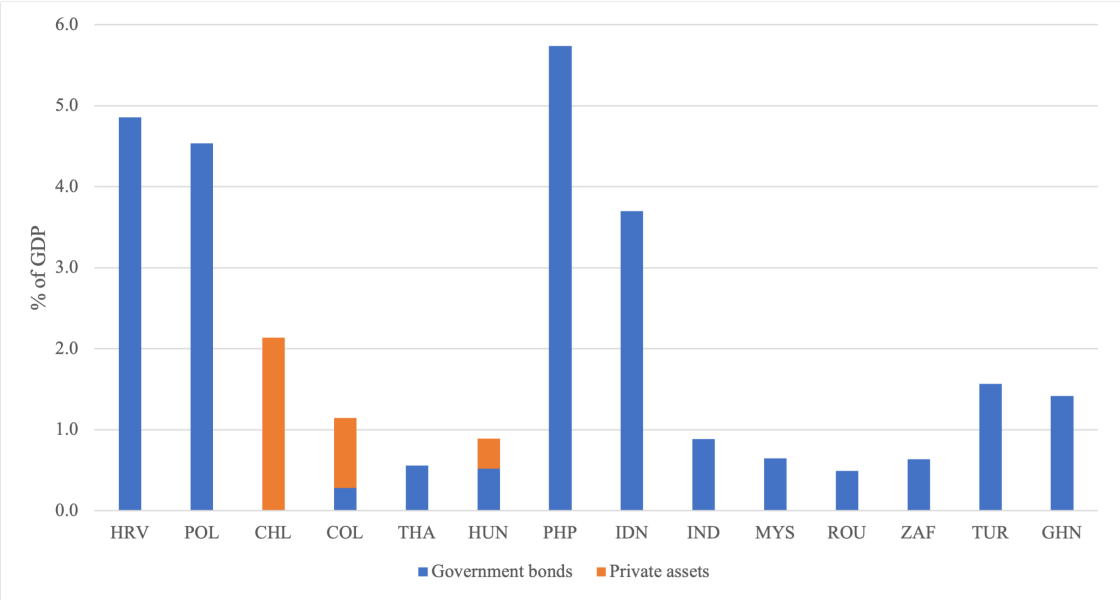

Figure 1: Asset purchases by EMDEs

Note: The vertical axis indicates the asset purchases as a percentage of 2020 GDP conducted from March to August in 2020. The horizontal axis shows the countries: Croatia (HRV), Poland (POL), Chile (CHL), Colombia (COL), Thailand (THA), Hungary (HUN), Philippines (PHP), Indonesia (IDN), India (IND), Malaysia (MYS), Romania (ROU), South Africa (ZAF), Turkey (TUR), and Ghana (GHN). The blue bar is the government bond purchase in the primary and secondary markets. The orange bar is the purchase of private sector assets including corporate bonds, asset-backed securities, exchange-traded fund, etc. *Source*: International Monetary Fund (2020).

These patterns motivate a theoretical framework built around two design choices for sterilized asset purchases. The first is the asset purchased: government bonds or corporate bonds, with the latter standing in for private asset purchases more broadly. The second is the sterilization instrument: FX reserve sales or domestic liability issuance, where the latter captures instruments such as security issuance and deposit facilities.

Overview of model The model builds on a standard small-open economy framework and incorporates a banking sector operating under financial frictions à la Gertler and Kiyotaki (2010) and Gertler and Karadi (2011), with flexible prices and incomplete financial markets. The model has three features that are key for the analysis. First, banks raise funds in foreign currency (i.e., in units of foreign goods) from households and foreign investors. This specification captures the stylized facts about *liability dollarization* in EMDEs.[3] Banks lend in domestic currency to non-financial firms and the government by purchasing their local-currency bonds (i.e., in units of domestic consumption). This funding intermediation activity makes the banks' balance sheet currency-mismatched, and hence, vulnerable to exchange rate depreciations. Second, banks operate under an occasionally-binding constraint imposed on the leverage, called the *leverage constraint*. Combined with liability dollarization, this friction generates the balance sheet effect, amplifying

[3]Despite the attempts to de-dollarize, many EMDEs still have a high degree of liability dollarization. See Yeyati (2021) for the recent trends of dollarization especially in Latin American countries. See also the growing literature on deposit dollarization, for example, Dalgic (2018), Christiano et al. (2021), and Ferrante and Gornemann (2022).

(a) Normal times

(b) Sudden stops

Figure 2: Flow of funds

Note: The figure describes the flow of funds in the economy in normal times (left panel) and in sudden stops (right panel). The arrow indicates the funding flow from the source to the recipient.

the impact of sudden stops. Third, in response to sudden stop, the government conducts asset purchase programs with *sterilization*. Specifically, in the baseline scenario, the government purchases corporate bonds, and at the same time, sells FX reserves. Furthermore, in using FX reserves for sterilization, the government faces an occasionally-binding constraint, called the *fear-of-losing-reserves constraint*.[4] When this constraint becomes binding, the government is balance-sheet constrained, and hence, can only purchase limited amounts of assets.

Figure 2 describes the flow of funds in the economy. In normal times (left panel), banks are at the center of funding flows, intermediating the supply of funds (from households and foreign investors) and the demand for funds (by non-financial firms and the government). The government in turn accumulates FX reserves from the international financial market. In sudden stops (right panel), however, it is the government that plays a key role in the funding intermediation. The government provides liquidity to domestic agents and resolves financial market disruptions through sterilized asset purchase programs.

Overveiw of results The first part of the paper studies the effectiveness of sterilized asset purchases in sudden stops. In a sudden stop, country interest rate rises and exchange rate depreciates, reducing the value of banks' net worth. Being constrained, banks are forced to decrease their asset demand, which causes a sharp fall in asset prices, deteriorating the value of net worth even further. This negative feedback loop is called the balance sheet effect that causes financial market disruptions. Sterilized asset purchase programs effectively break down the balance sheet effect. Specifically, the government asset purchases reduce banks' holdings of corporate bonds. These additional asset demands also prop up asset prices. The sales of FX reserves support exchange rates, reducing the repayment burdens on the foreign-currency liability. Hence, the policy enhances the value of banks' net worth, relaxing the leverage constraint. Consequently, the economy experiences a milder sudden stop, achieving welfare gains.

[4]This specification captures the "fear of losing reserves" in EMDEs. Aizenman and Sun (2012) finds that EMDEs depleted the accumulated reserves no more than 1/4 or 1/3 of the pre-crisis stock and rather let the exchange rate depreciate during the Global Financial Crisis.

The paper highlights policy trade-offs associated with asset purchases: deep recession versus slow recovery. Large-scale asset purchases buffer the economy from a deep recession; however, they slow down the subsequent recovery. The paper also emphasizes the role of FX reserves as a war chest. Since the policy room for asset purchases is limited by the stock of FX reserves, the government must hoard large enough FX reserves to have viable policy actions in sudden stops.

A counterfactual analysis of COVID-19 is conducted to assess the effectiveness of sterilized asset purchases. The analysis compares the economic outcomes under different policy scenarios. The baseline policy is compared with two counterfactuals: (i) *No policy*; and (ii) *Optimal policy* that maximizes welfare following the pre-specified rule.

The second part of the paper studies the design of sterilized asset purchase programs. Two alternative ways of conducting the policy are compared with the benchmark where the government purchases corporate bonds sterilized with FX reserves. One alternative is to purchase government bonds instead of corporate bonds. The other alternative is to issue government bonds, instead of selling FX reserves, to sterilize corporate bond purchases. It is shown that the benchmark policy is more effective in breaking down the balance sheet effect than the two alternatives.

Related literature This paper is related to the literature studying the role of financial frictions in amplifying external shocks. The seminal contributions of Krugman (1999), Aghion, Bacchetta and Banerjee (2004), Céspedes, Chang and Velasco (2004), and Gertler, Gilchrist and Natalucci (2007) show that financial market imperfection is the key to understand financial crises and the associated policies. Céspedes, Chang and Velasco (2004) show that currency mismatch gives rise to an amplification mechanism generating the balance sheet effect. Gertler, Gilchrist and Natalucci (2007) show that the financial accelerator mechanism can account for large contractions in financial crises like the 1997 Asian Financial Crisis. Closest in spirit to the policy setting of this paper is Burnside, Eichenbaum and Rebelo (2001), who show that unsterilized government interventions in the banking sector create inflation expectations that amplify crises. The present paper takes sterilization as given and instead focuses on the design and effectiveness of sterilized asset purchase programs. A similar modeling approach to banking sector frictions is taken by recent studies including Aoki, Benigno and Kiyotaki (2016), Kitano and Takaku (2020), and Akinci and Queralto (2022). The present paper allows the banks' leverage constraint to bind only occasionally, in particular, in a sudden stop. This specification of nonlinearity is similar to Karadi and Nakov (2021) that studies the optimal quantitative easing in a closed-economy context.[5] This paper finds a similar policy trade-off due to slow recovery, but through a newly documented channel arising from the currency-mismatched liability side of banks' balance sheet.

This paper relates to the literature on unconventional monetary policies in open economies. Jeanne and Korinek (2010), Bianchi (2011), and Bianchi and Mendoza (2018) propose the use of capital controls to correct the pecuniary externalities that induce inefficient private borrowings.[6] Céspedes, Chang and Velasco (2017) and Chang and Velasco (2017) analyze three unconventional measures: direct lending, liquidity facilities, and equity injections. They show that relaxing the financial constraint imposed on domestic banks

[5] See also Bocola (2016) and Akinci and Queralto (2022) for modeling works incorporating nonlinearity.
[6] See Bianchi and Mendoza (2020) and Erten, Korinek and Ocampo (2021) for a survey of the literature.

is the key for the policies to be effective. Chang (2018) further extends the analysis by examining FX interventions.[7] More recently, Adrian et al. (2020) and Basu et al. (2020) develop frameworks to guide the use of these unconventional tools to complement the conventional interest rate policy. Mimir and Sunel (2021) is closest to the present paper. While both papers study asset purchase programs in a similar framework, the present paper mainly focuses on programs sterilized through FX reserve sales. FX sales mitigate exchange rate depreciations, relaxing banks' leverage constraints more effectively. This paper also highlights the role of occasionally-binding constraints and provides welfare implications based on the policy trade-offs.

Lastly, this paper relates to the literature on the motives for hoarding FX reserves. In Jeanne and Ranciere (2011), FX reserves are insurance that reduces the probability of sudden stop occurrence. In Céspedes, Chang and Velasco (2017), Gopinath and Stein (2018), Bocola and Lorenzoni (2020), and Céspedes and Chang (2024), FX reserves are a war chest for the lender of last resort policy.[8] This paper emphasizes the war chest role of FX reserves. In particular, it shows that FX reserves can be used as a sterilization tool for asset purchase programs. By imposing an occasionally-binding constraint on the use of FX reserves, this paper emphasizes the importance of reserve accumulation in normal times before a sudden stop hits the economy.[9]

Outline The paper is organized as follows. Section 2 describes the model. Section 3 presents numerical results from the model simulations. Finally, Section 4 concludes.

2 Model

Consider a small-open economy with two tradable goods: domestically produced goods (home goods) and imported goods from the rest of the world (foreign goods). The domestic consumer price index (CPI) is defined as

$$P_t \equiv \left[(1-\gamma)P_{Ht}^{1-\eta} + \gamma P_{Ft}^{1-\eta}\right]^{\frac{1}{1-\eta}}$$

where P_{Ht} and P_{Ft} are the prices for home and foreign goods, γ is the trade openness, and η is the trade elasticity. Assume that the law of one price holds and that the weight of home goods in the consumption basket for the rest of the world is infinitely small. Under this assumption, the real exchange rate is expressed as

$$e_t = \frac{P_{Ft}}{P_t}$$

[7] See also the recent work on FX interventions along this line including Carrasco and Hoyle (2020) and Hofmann, Patel and Wu (2021).

[8] See also Das, Gopinath, Kim and Stein (forthcoming).

[9] Basu, Ghosh, Ostry and Winant (2018) study FX interventions in a stylized model with a similar constraint on the use of FX reserves. Kim (2025) studies a related reserve-management problem without commitment.

An increase in e_t indicates a real depreciation. The terms of trade is defined as

$$tot_t \equiv \frac{P_{Ft}}{P_{Ht}}$$

Let $p_{Ht} \equiv P_{Ht}/P_t$ be the relative price of home goods in terms of domestic CPI.

The economy is inhabited by households, banks, non-financial firms, and the consolidated government.

2.1 Households

There is a unit measure of households. Each household is composed of workers (fraction $1-\zeta$) and bankers (fraction ζ). In each period, a banker can operate the business with probability χ. Hence, $(1-\chi)\zeta$ of bankers terminates the business and becomes workers, in which case they give retained earnings as payouts to the households. To replace these exiting bankers, the same number of workers becomes new bankers, holding the relative proportion of workers and bankers fixed. The new bankers start the business with the net worth given as transfers by households. Finally, to maintain a representative-agent framework, assume that (i) workers put deposits in the banks owned by other households, and that (ii) within household, there is perfect consumption insurance.

A representative household has the GHH preference (Greenwood, Hercowitz and Huffman 1988)

$$U(c_t, h_t) = \frac{\left(c_t - \frac{\kappa_h}{1+\varphi} h_t^{1+\varphi}\right)^{1-\sigma} - 1}{1-\sigma}$$

where c_t is a constant elasticity of substitution (CES) composite of consumption on home and foreign goods,

$$c_t = \left[(1-\gamma)^{\frac{1}{\eta}} c_{Ht}^{\frac{\eta-1}{\eta}} + \gamma^{\frac{1}{\eta}} c_{Ft}^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}$$

h_t is labor hour, β is the discount factor, σ and φ are the inverse of the intertemporal elasticity of substitution and the Frish elasticity, and κ_h is the weight on labor disutility.[10]

The budget constraint expressed in real term is

$$c_t + e_t d_{Ht} = w_t h_t + e_t R_t d_{Ht-1} + \Pi_t - T_t \tag{1}$$

where d_{Ht} is a one-period riskless bank deposit in foreign currency (i.e., in units of foreign goods), R_t is a return on deposits, w_t is wage, Π_t is profits from banks and non-financial firms, and T_t is a lump-sum tax.

[10]The GHH preference is commonly used in the small-open economy literature (Neumeyer and Perri 2005; Uribe and Yue 2006; Mendoza 2010). It eliminates wealth effects on labor supply, preventing an artificial boom following a negative shock on the country interest rate.

Maximizing $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, h_t)$ subject to (1) gives the Euler equation and labor supply condition

$$1 = \mathbb{E}_t \left(\Lambda_{t,t+1} R_{t+1} \frac{e_{t+1}}{e_t} \right)$$
$$\kappa_h h_t^{\varphi} = w_t$$

where $\Lambda_{t,t+1} \equiv \beta \mathbb{E}_t (mu_{t+1}/mu_t)$ is the stochastic discount factor with $mu_t \equiv \left(c_t - \kappa_h h_t^{1+\varphi}/(1+\varphi) \right)^{-\sigma}$. The demands for home and foreign goods are $c_{Ht} = (1-\gamma)(p_{Ht})^{-\eta} c_t$ and $c_{Ft} = \gamma (e_t)^{-\eta} c_t$ from the standard intratemporal problem.

2.2 Banks

Banks raise funds with a one-period debt in foreign currency (i.e., in units of foreign goods). The total debt d_t is composed of households' deposit d_{Ht} and borrowing from foreign investors d_{Ft} (i.e., $d_t \equiv d_{Ht} + d_{Ft}$), both of which are subject to interest rate R_t given as exogenous.[11] Banks lend funds to both non-financial firms and the government by purchasing their bonds. Specifically, banks purchase corporate bonds s_{Bt} and government bonds b_{Bt} denominated in local currency (i.e., in units of domestic consumption). The corporate bond is a state-contingent claim on the future returns from one unit of physical capital of non-financial firms. Let q_{kt} and R_{kt} be the price and the return of a corporate bond. The government bond is a perpetuity that pays a fixed coupon payment each period. Let q_{bt} and R_{bt} be the price and the return of a government bond.

An individual bank $j \in [0, \zeta]$ has a balance sheet

$$q_{kt} s_{Bjt} + q_{bt} b_{Bjt} = n_{jt} + e_t d_{jt} \tag{2}$$

where n_{jt} is the net worth of bank j. Notice the mismatch of currency denomination between the assets and the liabilities on the balance sheet. The budget constraint is

$$e_t R_t d_{jt-1} + q_{kt} s_{Bjt} + q_{bt} b_{Bjt} = e_t d_{jt} + R_{kt} q_{kt-1} s_{Bjt-1} + R_{bt} q_{bt-1} b_{Bjt-1} \tag{3}$$

Combining (2) and (3),

$$n_{jt} = R_{kt} q_{kt-1} s_{Bjt-1} + R_{bt} q_{bt-1} b_{Bjt-1} - e_t R_t d_{jt-1} \tag{4}$$

indicating that the net worth is accumulated through retained earnings.

The bank's objective is to maximize the terminal payouts to the household in the form of net worth. The franchise value of the bank is

$$V_t \left(n_{jt} \right) = \max_{s_{Bjt}, b_{Bjt}, d_{jt}} \mathbb{E}_t \Lambda_{t,t+1} \left[(1-\chi) n_{jt+1} + \chi V_{t+1} \left(n_{jt+1} \right) \right] \tag{5}$$

[11] The full dollarization assumption is not crucial for the main results. See as a robustness check Appendix E.2 for partial dollarization.

taking into account the probability of terminating the business. Importantly, the bank is subject to the occasionally-binding incentive compatibility constraint given as

$$V_t(n_{jt}) \geq \theta(q_{kt}s_{Bjt} + \Delta q_{bt}b_{Bjt}) \tag{6}$$

where $\theta \in (0, 1)$ is the absconding rate of assets, and $\Delta \in [0, 1)$ denotes the relative absconding rate of the government bond compared to the corporate bond (i.e., the government bond is harder to be diverted).[12] This constraint gives rise to non-linear dynamics depending on whether it is binding or not.

The bank's problem is to maximize (5) subject to (2), (4), and (6).[13] The first-order conditions are

$$\mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}\left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \theta\lambda_t$$
$$\mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}\left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \Delta\theta\lambda_t$$

The left-hand sides are the expected excess returns on the corporate and the government bonds evaluated with the bank's stochastic discount factor $\Lambda_{t,t+1}\Omega_{t+1}$, while λ_t on the right-hand sides measures the tightness of constraint (6).[14]

When constraint (6) is not binding ($\lambda_t = 0$), $\Omega_{t+1} = 1$ and the standard no-arbitrage condition holds: excess returns on both bonds are zero and asset demand is independent of net worth. When it is binding ($\lambda_t > 0$), excess returns are positive and the first-order conditions reduce to the no-arbitrage condition between the two assets,

$$\Delta\mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}\left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}\left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right)$$

Define the risk-adjusted leverage ratio

$$l_{jt} \equiv \frac{q_{kt}s_{Bjt} + \Delta q_{bt}b_{Bjt}}{n_{jt}} \tag{7}$$

where Δ fraction of government bonds is included because the government bond is safer than the corporate bond by exactly the relative absconding rate Δ. Constraint (6) imposes a limit on this leverage ratio,

$$l_{jt} \leq \bar{l}_t \tag{8}$$

where $\bar{l}_t \equiv \frac{\mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}R_{t+1}e_{t+1}/e_t}{\theta - \mathbb{E}_t\Lambda_{t,t+1}\Omega_{t+1}(R_{kt+1} - R_{t+1}e_{t+1}/e_t)}$ depends only on aggregate variables. When binding, bank asset demand becomes proportional to net worth: $q_{kt}s_{Bjt} + \Delta q_{bt}b_{Bjt} = \bar{l}_t n_{jt}$, so a fall in net worth reduces

[12] As standard in the literature, the underlying assumption for this financial friction is that bankers can abscond with $\theta(q_{kt}s_{Bjt} + \Delta q_{bt}b_{Bjt})$ of assets. In such case, the banker exits the industry and the lenders (i.e., households and foreign investors) reclaim the rest of the assets. The given constraint restricts this possibility so that the bankers have no incentives to run away with the assets, and hence, the lenders are willing to lend for the bankers to operate in equilibrium.

[13] The details of bank's problem are given in Appendix A.

[14] As shown in Appendix A, $\Omega_t \equiv (1 - \chi) + \chi(\partial V_t(n_{jt})/\partial n_{jt})$ and λ_t is an increasing function of the Lagrange multiplier for constraint (6).

aggregate asset demand.

Aggregation Aggregate net worth is

$$n_t = \chi n_{ot} + (1-\chi) n_{yt} \tag{9}$$

where $n_{ot} = (R_{kt} - R_t e_t/e_{t-1}) q_{kt-1} s_{Bt-1} + (R_{bt} - R_t e_t/e_{t-1}) q_{bt-1} b_{Bt-1} + R_t (e_t/e_{t-1}) n_{t-1}$ is existing banks' retained earnings and $n_{yt} = \frac{\iota}{1-\chi}(q_{kt-1} s_{Bt-1} + \Delta q_{bt-1} b_{Bt-1})$ is the transfer to new banks. Assuming symmetric leverage ratios ($l_{jt} = l_t$ for all j), the aggregate leverage constraint is

$$l_t = \frac{q_{kt} s_{Bt} + \Delta q_{bt} b_{Bt}}{n_t} \leq \bar{l}_t \tag{10}$$

When binding, $q_{kt} s_{Bt} + \Delta q_{bt} b_{Bt} = \bar{l}_t n_t$, so a fall in aggregate net worth reduces aggregate asset demand.

Let μ_t denote the spread on the corporate bond

$$\mu_t \equiv \mathbb{E}_t (R_{kt+1} - R_{t+1} e_{t+1}/e_t)$$

Since it becomes positive when banks are constrained, μ_t serves as an indicator of financial market disruptions.

2.3 Non-financial firms

The non-financial firms are owned by households, operating in a perfectly competitive manner. The firms are composed of home goods producers, capital storage firms, and capital producers.

Home goods producers The home goods producers own Cobb-Douglas production technology

$$y_{Ht} = a_t k_{t-1}^{\alpha} h_t^{1-\alpha}$$

where α is the capital income share and a_t is the total factor productivity. To produce goods in period t, the home goods producers rent capital k_{t-1} from the capital storage firms at rate z_t and hire labor h_t at wage w_t. The first-order conditions from the profit-maximization problem are

$$z_t = \alpha p_{Ht} \frac{y_{Ht}}{k_{t-1}}$$
$$w_t = (1-\alpha) p_{Ht} \frac{y_{Ht}}{h_t}$$

Capital storage firms The capital storage firms purchase capital at price q_{kt-1}, rent it to home goods producers at rate z_t, and sell the undepreciated $(1-\delta)$ fraction at price q_{kt}. They finance capital purchases

by issuing corporate bonds to banks ($s_{t-1} = k_{t-1}$), paying return

$$R_{kt} = \frac{z_t + (1-\delta)q_{kt}}{q_{kt-1}}$$

Due to perfect competition and perfect monitoring by banks, capital price equals the bond price and firms earn zero profit.

Capital producers The capital producers produce new capital from investment subject to adjustment costs. Investment i_t is a CES composite of home and foreign goods defined analogously to consumption. Capital producers sell capital k_t to capital storage firms and repurchase undepreciated capital $(1-\delta)k_{t-1}$ at price q_{kt}. The problem is

$$\max_{\{k_t, i_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left[q_{kt}(k_t - (1-\delta)k_{t-1}) - i_t \right]$$

subject to

$$k_t = (1-\delta)k_{t-1} + i_t \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 \right]$$

where κ_i is a parameter for adjustment costs. The first-order condition is

$$1 = q_{kt} \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 - \kappa_i \left(\frac{i_t}{i_{t-1}} - 1 \right) \left(\frac{i_t}{i_{t-1}} \right) \right] + \kappa_i \mathbb{E}_t \Lambda_{t,t+1} q_{kt+1} \left(\frac{i_{t+1}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2$$

The investment for home and foreign goods are $i_{Ht} = (1-\gamma)(p_{Ht})^{-\eta} i_t$ and $i_{Ft} = \gamma(e_t)^{-\eta} i_t$ from the standard intratemporal problem similar to the one for consumption.

2.4 Government

The consolidated government makes a fixed public spending $g_t = g$ of home goods. It hoards FX reserves f_t purchased from the international financial market. FX reserves are one-period assets in foreign currency (i.e., in units of foreign goods) that pay return of R_t^*. The government also issues a long-term bond b_t at price q_{bt}, which is a perpetuity with geometrically decaying coupon payments. The return from holding the government bond from period $t-1$ to t is

$$R_{bt} = \frac{\Xi + \varrho q_{bt}}{q_{bt-1}} \tag{11}$$

where Ξ is the fixed coupon payment and ϱ is the decaying rate of the bond.

The balance sheet is

$$q_{kt} s_{Gt} + e_t f_t = q_{bt} b_t \tag{12}$$

The left-hand side is the value of total assets composed of corporate bonds s_{Gt} and FX reserves f_t, while the right-hand side is the value of net liabilities b_t.[15] The budget constraint is

$$p_{Ht}g_t + R_{bt}q_{bt-1}b_{t-1} + q_{kt}s_{Gt} + e_t f_t = q_{bt}b_t + R_{kt}q_{kt-1}s_{Gt-1} + e_t R_t^* f_{t-1} + T_t \tag{13}$$

Sterilized asset purchase Consider that the government conducts sterilized asset purchases. In particular, the government purchases corporate bonds and sterilizes with FX reserves. In the balance sheet, this operation involves adjustments in the asset composition

$$q_{kt} \underbrace{s_{Gt}}_{\uparrow} + e_t \underbrace{f_t}_{\downarrow} = q_{bt} \underbrace{b_t}_{\text{constant}}$$

holding the supply of government bond fixed $b_t = b$.[16] Assume that the government follows a policy rule to sell fraction Γ_t of FX reserves for sterilization. Hence, the remaining stock of FX reserves is

$$f_t = (1 - \Gamma_t)f \tag{14}$$

where f is the steady-state FX reserve holdings. Define the fraction Γ_t as

$$\Gamma_t \equiv \phi_\mu(\mu_t - \mu)$$

where $\mu_t - \mu$ is the difference between the spread on the corporate bond and its steady-state value and $\phi_\mu > 0$ is the degree of intervention. Since μ_t is considered as a measure of financial market disruptions, Γ_t is the extent of government intervention to stabilize the financial market. The higher ϕ_μ in Γ_t is, the more aggressive the government intervention is in response to the financial market disruptions measured by $\mu_t - \mu$.

Assume further a limit on the use of FX reserves

$$\Gamma_t \leq \bar{\Gamma}$$

where $\bar{\Gamma} \in (0, 1]$. Notice that this constraint, called fear-of-losing-reserves constraint, is occasionally-binding depending on the size of Γ_t. When it is binding, the government would not be able to purchase assets as much as it would without the constraint.

[15] Assume that the net worth of the government is constant and normalized to be zero, abstracting from the issues related to the evolution and management of government net worth.

[16] Section 3.5 discusses other ways of conducting sterilized asset purchases. For example, the government could retire the government bonds b_t instead of purchasing corporate bonds s_{Gt}. Also, the government could issue additional bonds b_t to sterilize asset purchases.

2.5 Rest of the model

The country interest rate is debt-elastic as in Schmitt-Grohé and Uribe (2003). It depends on the difference between the net foreign debt to GDP ratio from its steady-state value

$$R_t = R_t^* + \psi \left[\exp\left(\frac{e_t(d_{Ft} - f_t)}{y_t} - \frac{e(d_F - f)}{y} \right) - 1 \right] + \xi_t$$

where R_t^* is the foreign interest rate, $\psi > 0$ measures the responsiveness to net foreign debt to GDP ratio, $y_t \equiv p_{Ht} y_{Ht}$ is the GDP, and ξ_t is a risk premium shock following an AR(1) process $\xi_t = \rho_\xi \xi_{t-1} + \varepsilon_{\xi t}$ that initiates a sudden stop.

The market clearing condition for home goods requires

$$y_{Ht} = c_{Ht} + i_{Ht} + g_t + x_t \tag{15}$$

where the export x_t is defined as

$$x_t \equiv \gamma^* tot_t^\eta y_t^*$$

with foreign trade openness parameter γ^* and foreign output y_t^*.[17] The trade balance is defined as the difference between exports and imports,

$$tb_t \equiv p_{Ht} x_t - e_t m_t \tag{16}$$

where $m_t = c_{Ft} + i_{Ft}$ is the import of foreign goods. The balance of payments is

$$tb_t = -e_t(d_{Ft} - f_t) + e_t(R_t d_{Ft-1} - R_t^* f_{t-1})$$

describing the law of motion for the country's net foreign debt.[18] The current account is the sum of the trade balance and the net interests on the net foreign asset

$$ca_t \equiv tb_t - e_t[(R_t - 1)d_{Ft-1} - (R_t^* - 1)f_{t-1}]$$

which can be reduced to

$$ca_t = -e_t[(d_{Ft} - d_{Ft-1}) - (f_t - f_{t-1})]$$

the change in country's net foreign asset position. Finally, market clearing conditions for the financial

[17] I assume a symmetric CES composite for foreign consumption c_t^* and investment i_t^* with trade openness γ^* and trade elasticity η. The foreign demand for home goods is $x_t = \gamma^* \left(\frac{p_{Ht}}{e_t}\right)^{-\eta} y_t^*$ where $y_t^* = c_t^* + i_t^*$.

[18] See Appendix B for the derivation.

markets are

$$s_t = s_{Bt} + s_{Gt}$$
$$b_t = b_{Bt}$$

where the left-hand (right-hand) sides indicate the supply of (the demand for) the corporate and the government bonds.

Equilibrium A competitive equilibrium is a set of processes $\{c_t, c_{Ht}, c_{Ft}, h_t, d_{Ht}, \Lambda_{t,t+1}, d_t, d_{Ft}, s_{Bt}, b_{Bt}, n_t, l_t, \lambda_t, \Omega_t, y_t, y_{Ht}, k_{t-1}, i_t, i_{Ht}, i_{Ft} b_t, s_{Gt}, f_t, g_t, T_t, \mu_t, \Gamma_t, x_t, m_t, tb_t, ca_t, p_{Ht}, e_t, tot_t, w_t, z_t, R_t, R_{kt}, R_{bt}, q_{kt}, q_{bt}\}$ such that (i) households, banks, and non-financial firms optimize under the constraints, and (ii) goods and financial markets clear, given the exogenous processes $\{a_t, y_t^*, R_t^*, \xi_t\}$. The complete equilibrium conditions are given in Appendix C.

3 Results

3.1 Parameterization and solution method

Table 1 summarizes the parameter values used in model simulation. Among others, $\{\beta, \sigma, \alpha, \delta, \eta_i\}$ is the set of parameters pinned down by the standard values in the literature. I set the inverse of Frisch elasticity to 1/3.[19] The trade elasticity is set to 1.5 as in Kitano and Takaku (2020).[20]

For calibration, I use yearly data from 2000 to 2020 for five countries: Chile, Colombia, Korea, Philippines, Turkey.[21] From the data, I compute the average values for government expenditure to GDP ratio (0.13), export to GDP ratio (0.26), external debt to GDP ratio (0.38), and FX reserve to GDP ratio (0.16). To compute the spread on the corporate and the government bonds in the steady state, I use lending rates, deposit rates, and 10-year government bond yields. The difference between lending rates and deposit rates (410 basis points per annum) captures the corporate bond spread in the model. The spread on government bond in the model is captured by the difference between 10-year government bond yields and deposit rates (150 basis points per annum).[22] The leverage ratio is calculated as banks' asset to capital ratio. As in Akinci and Queralto (2022), I define banks' capital as the sum of total capital and other liabilities. The average leverage ratio is 6.3 for the five countries.[23]

The calibrated parameters are as follows. The utility weight on labor is set for the steady-state labor

[19] The conventional range for the Frisch elasticity is between 1 and 3 in the literature. As in Gertler and Kiyotaki (2010), I set the Frisch elasticity to a relatively high value to reflect other frictions absent in the model (e.g. nominal rigidities).

[20] This value can also be found in the early international real business cycle literature (e.g., see Backus, Kehoe and Kydland 1994).

[21] See Appendix D for the data sources.

[22] The lending rate is the interest rate charged by banks for the short-term and medium-term financing needs of the private sector. The deposit rate is the interest rate offered by banks for demand, time, or savings deposits. See for details the introductory notes for the International Financial Statistics at the IMF.

[23] This value falls into the conventional range in the literature. While it is set to 4 in Gertler and Kiyotaki (2010) and Gertler and Karadi (2011), the leverage ratio in open economy models typically has a higher values ranged from 6 to 7 (e.g., see Mimir and Sunel 2021 and Akinci and Queralto 2022).

Parameter	Value	Description	Source/Target
β	0.99	Discount factor	Standard
σ	2	Inverse of intertemporal elasticity of substitution	Standard
α	0.33	Income share for capital	Standard
δ	0.025	Depreciation rate	Standard
κ_i	1	Adjustment cost in capital production	Standard
κ_h	2.2434	Utility weight on labor	Labor hours
φ	1/3	Inverse of Frisch elasticity	Gertler and Kiyotaki (2010)
η	1.5	Trade elasticity	Kitano and Takaku (2020)
γ	0.29	Trade openness	Export-GDP ratio
Δ	0.37	Relative divertible fraction of gov. bond	Spread on gov. bond
χ	0.92	Banks' surviving probability	Mimir and Sunel (2021)
ι	{0.0112,0.0019}	Transfer rate for new banks	Small positive number
θ	{0.1587,0.6116}	Divertible fraction of total assets	Leverage ratio and spreads
Ξ	0.0275	Coupon payment for gov. bond	Spread on gov. bond
ϱ	0.9848	Decaying rate for gov bond	10 years duration
g	0.13	Government public spending	Gov. expenditure to GDP ratio
f	0.64	FX reserves	Reserve to GDP ratio (annual)
ϕ_μ	{250,1000,2500}	Responsiveness to spreads in policy rule	
$\bar{\Gamma}$	{1,0.3}	Limit on the use of FX reserves	
ρ_ξ	0.91	Persistence in risk premium	AR(1) estimation
ψ	0.001	Elasticity of debt in interest rate	Small positive number
R^*	1.0101	Foreign interest rate	Implied from model
y^*	1	Foreign output	Normalization
γ^*	0.344	Foreign trade openness	Implied from model

Table 1: Parameter values

Note: The table summarizes the parameter values used in the simulations. There are two sets of values for θ and ι. For the analysis in Sections 3.2, 3.3 and 3.4, θ and ι are set to 0.0112 and 0.1587. In Section 3.5, θ and ι are set to 0.0019 and 0.6116. For the parameters related to the policy rule, I consider three values of ϕ_μ and two values of $\bar{\Gamma}$ as shown in the table.

hours to be 1/3. The trade openness is 0.29 to target the export to GDP ratio. The surviving probability for banks is set to 0.92, as in Mimir and Sunel (2021), targeting the 3-year average life of banks. The parameters for the government bond $\{\Xi, \varrho\}$ are set to target the spread and 10-year duration of the government bond. The parameter set $\{\theta, \iota, \Delta\}$ targets the corporate and the government bond spreads, and the leverage ratio just equal to its maximum value. For the analysis in Sections 3.2 and 3.3, I set θ and ι to 0.0112 and 0.1587 for zero spreads on the bonds (i.e., $\mu = 0$) so that banks are not constrained in the steady state. In Section 3.5, θ and ι are set to 0.0019 and 0.6116 to analyze the equilibrium around the steady state where banks are constrained (i.e., $\mu > 0$). Finally, for the parameters related to the policy rule, I consider three values of ϕ_μ and two values of $\bar{\Gamma}$ as shown in the table.

Turning to the parameters for the country interest rate and the foreign variables, the persistence parameter for risk premium is set to 0.91, which is the average of estimates from fitting an AR(1) process on J.P. Morgan Emerging Markets Bond Spread (EMBI+) series in quarterly frequency.[24] Following Schmitt-Grohé and Uribe (2003), I set the debt elasticity parameter in the interest rate to a small positive number so that it has little impact on the results but help make the model stationary. Lastly, the steady-state foreign output is normalized to 1, the steady-state foreign interest rate and foreign trade openness are implied from the equilibrium conditions in the steady state.

Solution method The model features two occasionally-binding constraints: the leverage constraint and the fear-of-losing-reserves constraint. In order to solve the model with these occasionally-binding constraints, I use the Levenberg-Marquardt mixed complementarity problem (LMMCP) solver by Kanzow and Petra (2004) incorporated in the perfect-foresight solver of Dynare package (Adjemian et al. 2022).[25] Using this method, I derive impulse responses of the endogenous variables to an unexpected shock in the initial period starting from the steady state. Besides the shock in the initial period, all future uncertainty is assumed to be anticipated. It is also assumed that the model economy moves back to the steady state within a finite horizon. The advantage of this solution method is that the model is solved exactly up to rounding errors fully taking into account model nonlinearities.

Model fit Table 2 examines performance of the model in terms of untargeted moments over business cycles. In particular, the table describes the correlations of consumption, investment, trade balance, current account, real exchange rates, banks' leverage, corporate bond spread with GDP. The table also reports the standard deviations of consumption and investment relative to the GDP. The model moments are computed from the simulation without policy under the risk premium shock. The first (second) column corresponds to the simulation starting from the steady state where banks' leverage constraint is not binding (binding). The data moments are from quarterly data for the five countries from 2000Q1 to 2020Q4.[26] Despite

[24] The quarterly series is constructed by averaging the original monthly series obtained from the Global Economic Monitor at the World Bank.

[25] The LMMCP uses the Newton method with a generalized Jacobian to solve simultaneously the equilibrium conditions stacked over the simulation periods. Karadi and Nakov (2021) also use this method to solve the model with occasionally-binding constraints. See Swarbrick (2021) for a review of solution methods dealing with occasionally-binding constraints.

[26] The variables y, c, and i are HP-filtered in logs. The variable ca is HP-filtered in levels. The trade balance is the difference between exports and imports HP-filtered in logs. See Appendix D for the data sources.

Moments	Model		Data		
			Average	Minimum	Maximum
$\rho(c, y)$	1.00	0.99	0.71	0.22	0.94
$\rho(i, y)$	0.85	0.79	0.49	0.34	0.75
$\rho(tb, y)$	-0.68	-0.61	-0.46	-0.66	-0.27
$\rho(ca, y)$	-0.66	-0.55	-0.41	-0.73	-0.22
$\rho(e, y)$	-0.3	-0.15	-0.15	-0.48	0.51
$\rho(l, y)$	-0.47	-0.53	-0.06	-0.17	0.11
$\rho(\mu, y)$	-0.51	-0.61	-0.16	-0.7	0.36
$\sigma(c)/\sigma(y)$	0.87	0.92	1.04	0.24	1.63
$\sigma(i)/\sigma(y)$	5.41	6.19	4.07	2.28	7.72

Table 2: Untargeted moments

Note: The table describes untargeted moments over business cycles. $\rho(\cdot, y)$ is the correlation with GDP. $\sigma(\cdot)/\sigma(y)$ is the standard deviation relative to GDP. The moments from the model are listed in the first two columns. The corresponding data moments are listed in the rest of the columns in terms of average, maximum and minimum values for the five countries.

incorporating only one shock, the model correctly captures the dynamics of business cycles for the five countries. Although the magnitude is greater, the model delivers the correct cyclicality. It reports strong procyclicality of consumption and investment. It also displays countercyclical trade balance, current account, and real exchange rates. The banks' leverage and the spread on corporate bond are countercyclical. The model delivers relative volatilities of consumption and investment reasonably close to the ones from data.

3.2 Sudden stop episode

Consider the steady state in which banks are not constrained (i.e., $\mu = 0$). Suppose there is an unanticipated rise in risk premium by 0.1 percentage point, initiating a sudden stop. This section analyzes the impact of a sudden stop on the economy with and without sterilized asset purchase programs.

Sudden stop without policy Consider first the case where the government does not conduct sterilized asset purchase programs. Figures 3 and 4 describe the responses of the key variables to the given risk premium shock. Figure 3 shows that, as the shock hits, the country interest rate rises (bottom-right panel). The economy experiences abrupt capital outflows (bottom-left panel) and exchange rate depreciation (bottom-middle panel). In terms of real activity (top panels), the sudden stop induces sharp hump-shaped contractions in output, investment, and consumption. The economy moves back to the steady state as the country interest rate goes down. What is the channel through which the sudden stop negatively affects the economy? How is the financial shock of risk premium transmitted to the real economy?

Figure 4 illustrates the responses of the variables related to financial markets. It shows that banks get constrained (bottom-left panel) in the first nine periods, which is mirrored by the positive spread on the corporate bond (top-right panel). The figure also shows that the banks' net worth falls sharply (top-left

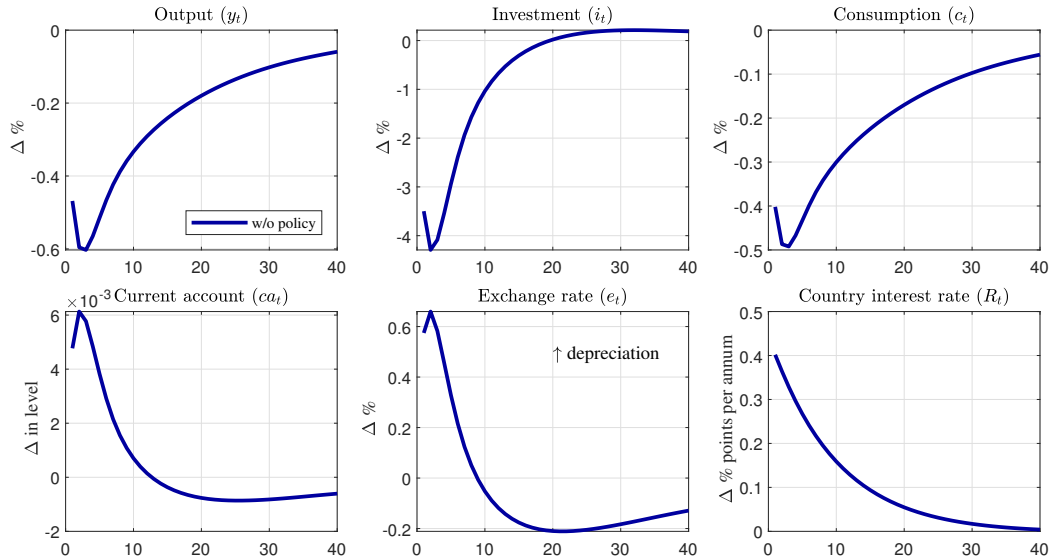

Figure 3: Impulse responses in a sudden stop without policy

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The horizontal axis indicates time periods in quarters.

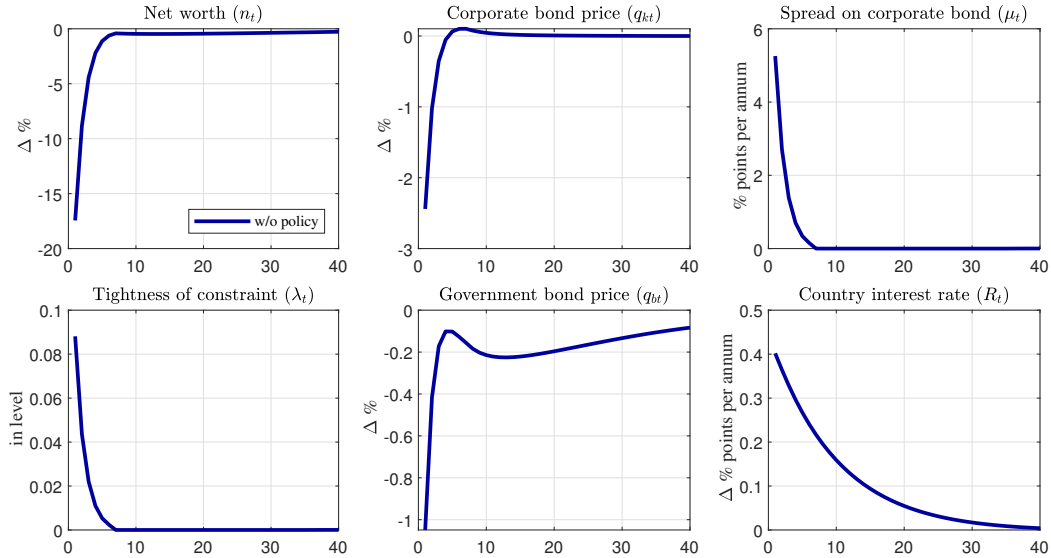

Figure 4: Impulse responses in a sudden stop without policy

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The horizontal axis indicates time periods in quarters.

panel). To examine the mechanism in more detail, recall that banks' net worth is the retained earnings

$$\underbrace{n_t}_{\downarrow} = R_{kt} q_{kt-1} s_{Bt-1} + R_{bt} q_{bt-1} b_{Bt-1} - \underbrace{e_t R_t d_{t-1}}_{\uparrow}$$

As the country interest rate and the exchange rate rise, the banks' net worth falls due to an increase in the value of liabilities. As the net worth falls, the constrained banks decrease their asset demand, reducing asset prices for the corporate and the government bonds (top- and bottom-middle panels). The reduction in asset prices brings down the value of net worth even further, creating a negative feedback loop. To see this, rewrite the net worth in terms of the current asset prices

$$\underbrace{n_t}_{\downarrow\downarrow} = \underbrace{(z_t + (1-\delta) q_{kt}) s_{Bt-1}}_{\downarrow} + \underbrace{(\Xi + \varrho q_{bt}) b_{Bt-1}}_{\downarrow} - \underbrace{e_t R_t d_{t-1}}_{\uparrow}$$

by replacing R_{kt} and R_{bt}. As the asset prices fall, the value of banks' asset holdings decreases, deteriorating the net worth. Consequently, banks lose more than 15 percent of the net worth on the onset of the sudden stop. Through this negative feedback loop, called the balance sheet effect, the sudden stop brings a severe disruption in the financial markets.

The financial market disruptions, in turn, spill over to the real economy. Specifically, as banks are constrained, the spread on corporate bonds rises, implying that non-financial firms face a high cost of credit. Since the firms are not able to borrow from the banks, their investment declines significantly, and thus, the economy suffers large contractions in real activity. The fall in aggregate demand for home goods causes the exchange rate to depreciate.

To sum up, the sudden stop brings a severe recession to the economy. The key amplification mechanism is the balance sheet effect that disrupts banks' funding intermediation activities.[27] Thus, it is crucial for the government to break down the balance sheet effect in order to buffer the economy against the sudden stop.

Sudden stop with sterilized asset purchases Suppose now that the government conducts sterilized asset purchase programs. Consider the same magnitude of risk premium shock that hits the economy in the same steady state. Figures 5, 6, and 7 describe impulse responses in the sudden stop episode. The blue solid lines are the responses without the policy as in the previous analysis. The red lines are the responses under sterilized asset purchase programs with three different degrees of government intervention $\phi_\mu = 250, 1000, 2500$.

Figure 5 shows that, under sterilized asset purchase programs, the impact of a sudden stop is significantly reduced. The economy experiences less capital outflows (bottom-left panel) and mild exchange rate depreciations (bottom-middle panel). Compared to the case without the policy, real activity suffers far less of a contraction (top panels).

[27] In the absence of the balance sheet effect, the impact of a sudden stop would be much milder. See Appendix E.1 for a counterfactual analysis where the leverage constraint is not imposed.

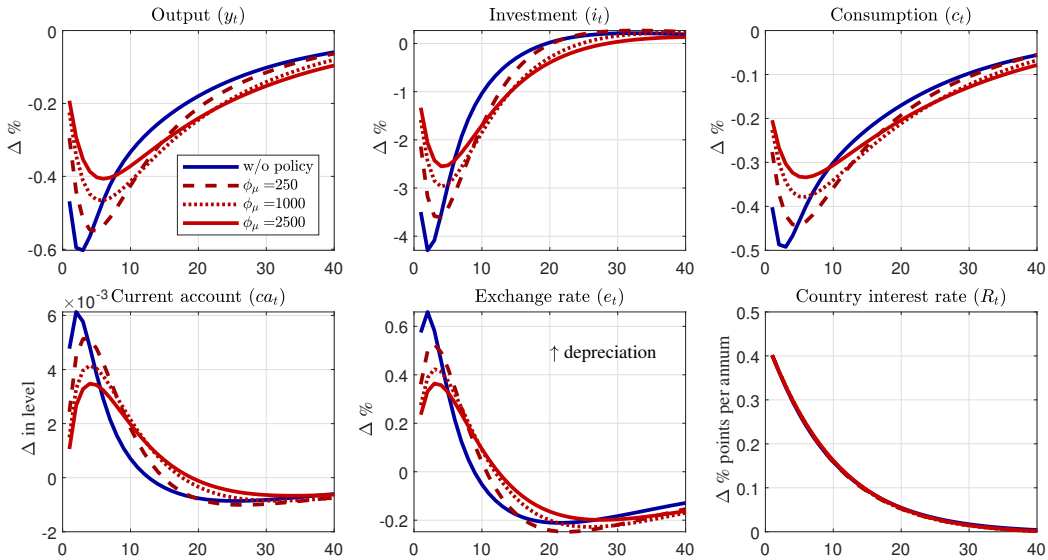

Figure 5: Impulse responses in a sudden stop with policy

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The blue solid lines are the responses without policy. The red lines are the responses under policy with ϕ_μ = 250, 1000, 2500. The horizontal axis indicates time periods in quarters.

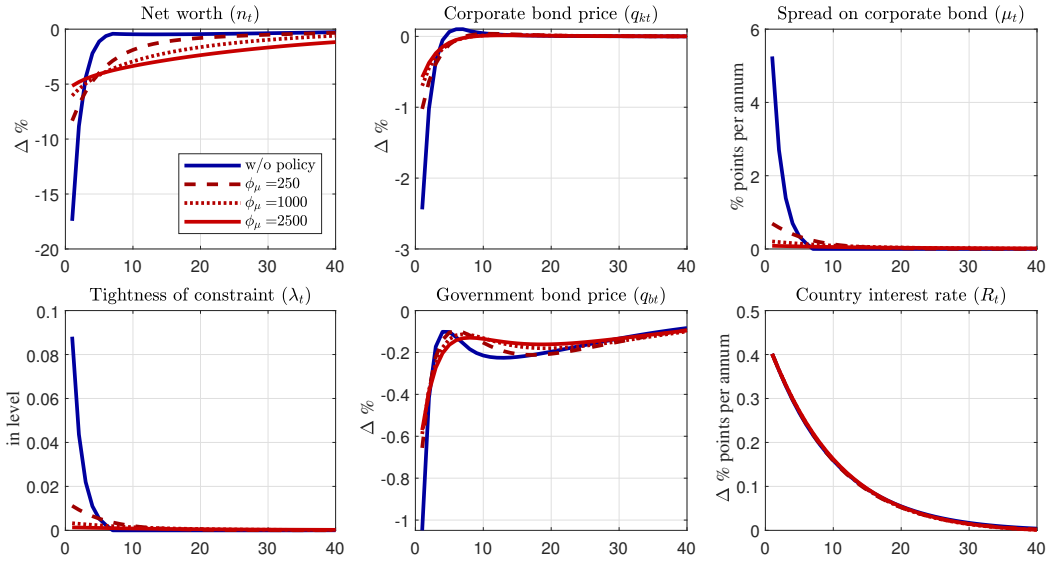

Figure 6: Impulse responses in a sudden stop with policy

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The blue solid lines are the responses without policy. The red lines are the responses under policy with ϕ_μ = 250, 1000, 2500. The horizontal axis indicates time periods in quarters.

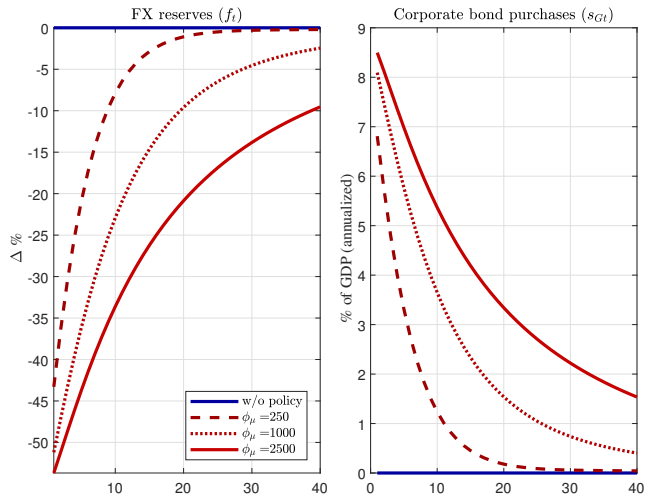

Figure 7: Impulse responses in a sudden stop with policy

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The blue solid lines are the responses without policy. The red lines are the responses under policy with $\phi_\mu = 250, 1000, 2500$. The horizontal axis indicates time periods in quarters.

Figure 6 describes how the policy eases financial market disruptions caused by the sudden stop. By purchasing corporate bonds, the government provides liquidity directly to non-financial firms. The government asset purchases also help banks relax the leverage constraint. Holding the supply fixed, it props up asset prices (top- and bottom-middle panels) and reduces banks' holdings of corporate bonds. Since the government sells FX reserves for sterilization, it supports exchange rates. Thus, the policy strengthens the value of banks' net worth (top-left panel). Consequently, the government breaks down the balance sheet effect, mitigating the impact of a sudden stop. Under the policy, banks are not as constrained as in the case without the policy (bottom-left panel). The lower spreads on the corporate bond (top-right panel) imply that non-financial firms are facing a relatively lower cost of credit.

Policy trade-offs Figure 7 shows that, when the government intervenes aggressively, the economy gets into a less severe recession. When it intervenes with a high ϕ_μ, the government purchases a large amount of corporate bonds (right panel). To sterilize these asset purchases, the government sells a large amount of FX reserves (left panel). For instance, under the most aggressive policy with $\phi_\mu = 2500$, the government sells more than 50 percent of FX reserve holdings and purchases corporate bonds more than 8 percent of GDP.

The benefit of large-scale asset purchase programs comes with a cost of slow recovery. The top-left panel of Figure 5 displays policy trade-offs in terms of output. The most aggressive policy with $\phi_\mu = 2500$ significantly dampens the impact of a sudden stop. The decline in output on impact is less than half of the one without the policy. Within the first 15 quarters, the contraction is much smaller compared to the ones under other less aggressive policies with $\phi_\mu = 250, 1000$. However, under this most aggressive policy, the output goes back to the steady state more slowly in the phase of recovery. Similar patterns can be observed

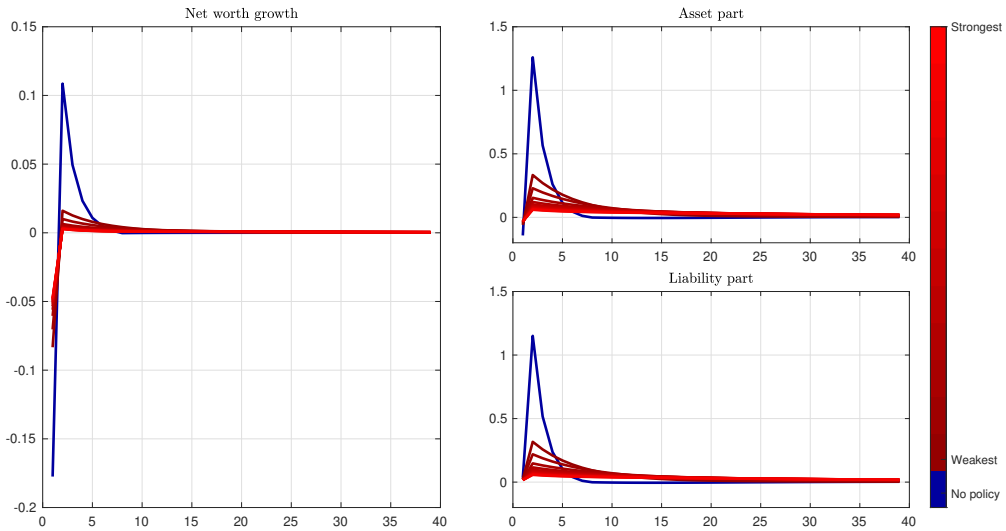

Figure 8: Net worth growth and its asset and liability parts

Note: The figure describes the growth of net worth for the banks relative to the steady-state value. The overall growth (left) is decomposed into the asset part (right top) and the liability part (right bottom). It is derived from the impulse response to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The vertical axis indicates the decimal growth rate. The horizontal axis indicates time periods in quarters. The colorbar indicates the degree of intervention corresponding to $\phi_\mu = 0$ (*No policy*), and ϕ_μ from 250 (*Weakest policy*) to 5000 (*Strongest policy*).

in the responses of investment and consumption.[28]

The slow recovery of the economy is caused by slow recapitalization of banks. To see this in more detail, note that the growth of banks' net worth can be rewritten as

$$\frac{n_t}{n_{t-1}} = \underbrace{(\chi R_{kt} + \iota)\left(\frac{q_{kt-1}s_{Bt-1}}{n_{t-1}}\right) + (\chi R_{bt} + \iota\Delta)\left(\frac{q_{bt-1}b_{Bt-1}}{n_{t-1}}\right)}_{\text{Asset part}} - \underbrace{R_t\frac{e_t}{e_{t-1}}\left(\frac{e_{t-1}d_{t-1}}{n_{t-1}}\right)}_{\text{Liability part}}$$

stating that net worth growth in period t is determined by the difference between returns on asset holdings (asset part) and repayments of liabilities (liability part) given the pre-determined asset-liability structure of the balance sheet in period $t-1$.

Figure 8 illustrates dynamics of banks' net worth growth (left panel) decomposed into the asset part (right top panel) and the liability part (right bottom panel). It is easy to see that the initial negative impact on net worth gets smaller when the policy intervention becomes stronger (in brighter red). In the recovery process, however, the opposite is true. On the one hand, stronger interventions support asset prices compressing banks' asset returns. On the other hand, stronger interventions reduce liability burdens via exchange rate support. Combining the two competing effects, net worth growth is lower with stronger interventions leading to slower recovery (i.e., decline in liability burdens does not fully offset reduced asset excess returns).[29]

[28] See Appendix E.3 for the marginal changes in FX reserve sales, corporate bond spreads, and net worth growth to the additional asset purchases.

[29] Karadi and Nakov (2021) also find that the stronger the policy intervention, the slower the recovery in a closed-economy context. The new channel documented in this paper is the additional margin interacted with exchange rates coming from the

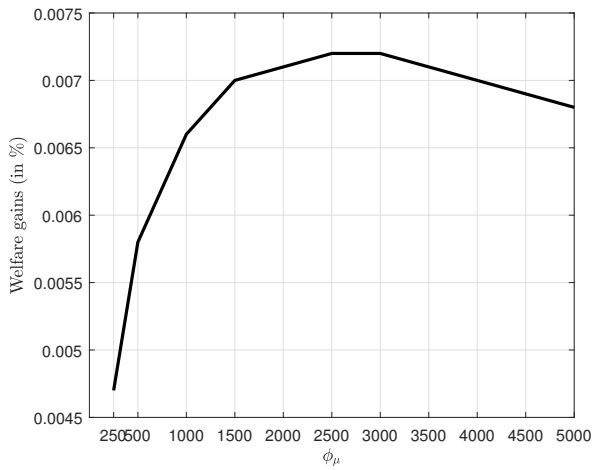

Figure 9: Welfare gains from asset purchase program

Note: The figure describes consumption-equivalent welfare gains as a function of the policy responsiveness parameter ϕ_μ.

Welfare analysis Does the sterilized asset purchase program bring welfare gains? In particular, considering the policy trade-offs, what is the degree of intervention that brings the largest welfare gains? To answer these questions, I examine the consumption-equivalent welfare measure. This welfare measure is defined as the fraction of consumption that households would give up so that they feel indifferent to live without the policy. The welfare measure ω is defined implicitly as

$$\sum_{t=0}^{\infty} \beta^t U(c_t^n, h_t^n) = \sum_{t=0}^{\infty} \beta^t U((1-\omega)c_t^p, h_t^p) \tag{17}$$

where c_t^n and h_t^n (c_t^p and h_t^p) are consumption and labor hours in equilibrium without (with) the policy.

Figure 9 plots the welfare gains as a function of the degree of intervention. The figure shows that the sterilized asset purchase program improves welfare. Compared to the economy without the policy, households enjoy nontrivial positive welfare gains. The welfare gain is the highest (0.0072%) at ϕ_μ = 2500. This means that households would be willing to permanently sacrifice 0.0072% of their equilibrium consumption to live in the economy where the government conducts sterilized asset purchases with ϕ_μ = 2500. The concave shape of welfare gains demonstrates the policy trade-offs. As the government intervenes more aggressively, welfare gain goes up until ϕ_μ = 2500. After this point, welfare gain diminishes as the degree of intervention increases. This means that, after the peak at ϕ_μ = 2500, the marginal cost of slow recovery is larger than the marginal benefit of buffering against the initial impact of a sudden stop.

3.3 The role of FX reserves in sterilized asset purchase program

To examine the role of FX reserves in more detail, suppose that the government conducts sterilized asset purchase programs under a more stringent constraint.

Specifically, assume that the government can only use up to 30 percent of FX reserves (i.e., $\bar{\Gamma}$ = 0.3).[30]

currency-mismatched liability of banks' balance sheet.

[30]This assumption is based on the empirical evidence in Aizenman and Sun (2012).

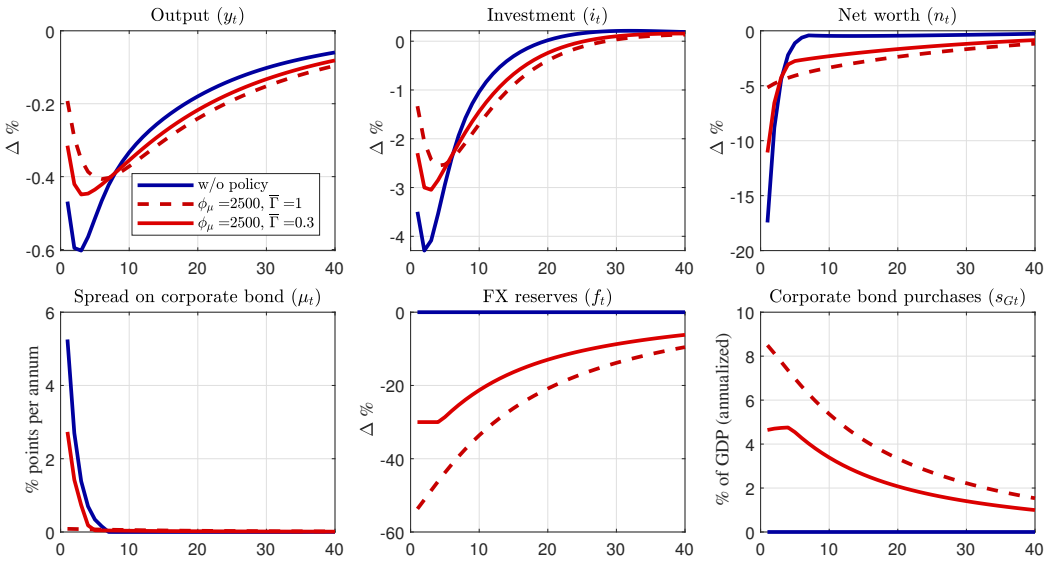

Figure 10: Impulse responses in a sudden stop with policy under constraint

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The blue solid lines are the responses without policy. The red lines are the responses under policy with ϕ_μ = 2500 and $\bar{\Gamma}$ = 1, 0.3. The horizontal axis indicates time periods in quarters.

Figure 10 describes the impulse responses in the same sudden stop episode. Again, the blue solid lines are the responses without the policy. The red dotted lines are the responses under the policy with ϕ_μ = 2500 as in the previous section. The red solid lines are the responses under the policy with the same degree of intervention but with the stringent constraint on FX reserves. As the figure shows, on the onset of sudden stop, the government uses FX reserves up to the limit (bottom-middle panel) to sterilize corporate bond purchases. Due to the more stringent constraint, asset purchases are limited up to about 5 percent of GDP (bottom-right panel). This is about a half of purchases that the government would make in the baseline analysis. Hence, the government cannot fully ease financial market disruptions as in the baseline. The fall in banks' net worth is about 10 percent (top-right panel) and the spread on the corporate bond rises to 3 percentage points (bottom-left panel). As a result, output and investment decrease more.

Suppose further that the government hoards small amounts of FX reserves. In particular, the government is facing the same constraint but only with a half of the baseline FX reserves. Figure 11 plots the impulse responses under this additional specification in the same sudden stop episode. The blue solid lines are the responses without the policy as before. The red solid lines are the responses with the policy under the constraint on FX reserves from Figure 10. The gray dotted lines are the responses with the policy under the same constraint but only with a half of FX reserves. As before, the use of FX reserves in asset purchases is limited up to 30 percent (bottom-middle panel). With only a half of FX reserves, the government can purchase corporate bonds about 2 percent of GDP (bottom-right panel). Banks lose the value of net worth by about 15 percent and the spread on the corporate bond rises to 4 percent. In terms of real activity, output and investment contract even more.

The analysis given in this section highlights the role of FX reserves in conducting sterilized asset purchases. When the government is constrained in using FX reserves, the policy space for asset purchases

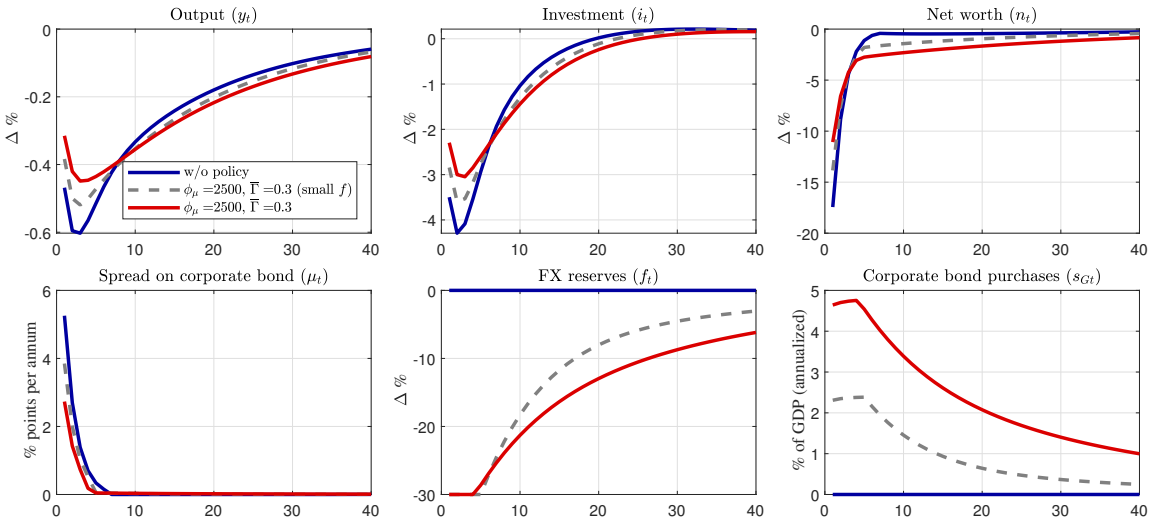

Figure 11: Impulse responses in a sudden stop with policy under constraint

Note: The figure describes the impulse responses to 0.1 percentage point of risk premium shock from the steady state where banks are not constrained. The blue solid lines are the responses without policy. The red solid and gray dotted lines are the responses under policy with ϕ_μ = 2500 and $\bar{\Gamma}$ = 0.3. In the gray dotted responses, the amount of FX reserves is only half of the amount in the red solid responses. The horizontal axis indicates time periods in quarters.

is limited. This implies that it is important to hoard enough ready-to-use FX reserves as a war chest to deal with sudden stops effectively.

3.4 Counterfactual analysis

This section provides a counterfactual analysis of the COVID-19 crisis to examine the effectiveness of sterilized asset purchases.

Calibration of risk premium shock Let μ^{data} denote the unexpected increase in average lending–deposit rate spreads in 2020Q2 for the five countries that the model is calibrated to. To calibrate the size of risk premium shock, define $\xi_{t=1}$ implicitly as below.

$$\hat{\mu}^{\text{data}} = \mu_p^{\text{model}}(\xi_{t=1}) \tag{18}$$

where $\hat{\mu}^{\text{data}}$ is the residual from applying an AR(1) process to the lending–deposit rate spreads, and $\mu_p^{\text{model}}(\xi_{t=1})$ is the model-implied spread measured on impact of the shock $\xi_{t=1}$ under the policy specification p.[31]

Consider as the baseline case with the *Baseline policy* where the government implements sterilized asset purchases with FX reserves following the policy rule specified in equation (14). Given this baseline policy specification, the scalar $\xi_{t=1}$ is found numerically from equation (18) by a one-dimensional root search. I simulate the following two counterfactuals and compare with the baseline. (i) *No policy*: the absence of policy interventions; (ii) *Optimal policy*: the presence of sterilized asset purchases to maximize

[31]Note that the estimate $\hat{\mu}^{\text{data}}$ is resulted from not only the unanticipated risk premium shock but also the policy responses. As discussed in Section 1, EMDEs implemented asset purchase programs for the first time during COVID-19.

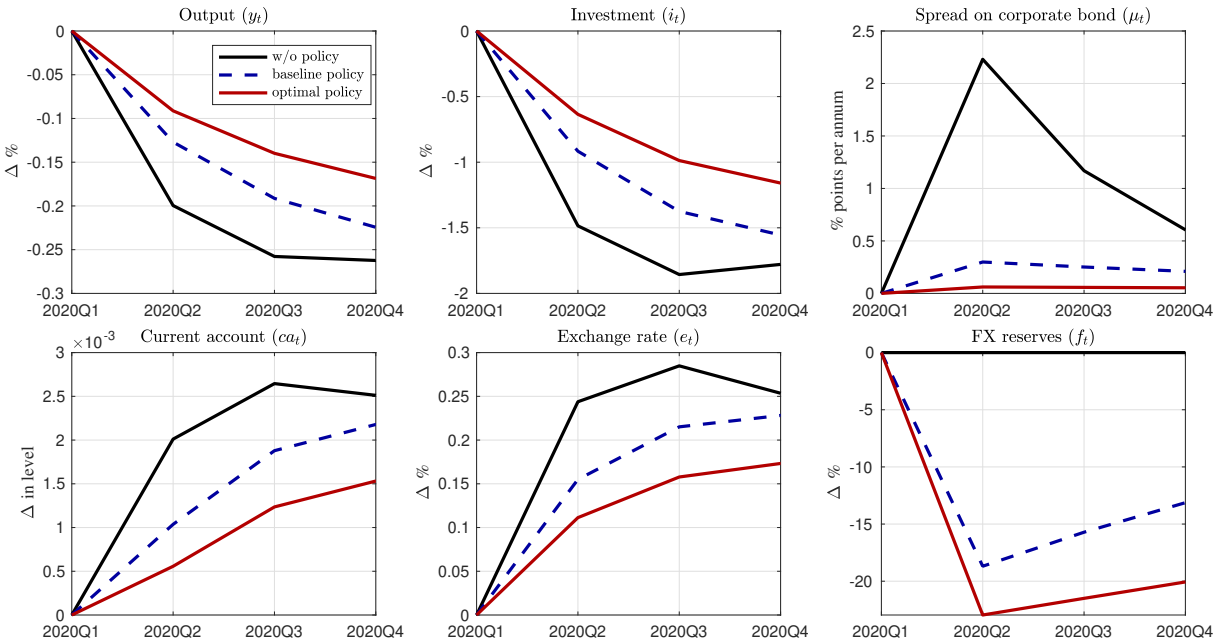

Figure 12: Counterfactual analysis of the COVID-19

Note: The figure describes the counterfactual analysis of the COVID-19 crisis. The black solid lines are the responses without policy interventions (*No policy*). The red solid lines are the responses under optimal sterilized asset purchases (*Optimal policy*). The blue dashed lines are the responses under the baseline case.

welfare given in (17).[32]

Figure 12 illustrates the two counterfactuals along with the baseline case. For the same COVID-19 risk premium shock, compared to the baseline, the contractionary effect is larger in the *No policy* case, while it is smaller in the *Optimal policy* case. In particular, the *No policy* case displays the largest spike in the corporate bond spread and the strongest exchange rate depreciation along with the largest current account reversal. The *Optimal policy* case substantially mutes contractionary effects even more so than the baseline while using FX reserves more intensively. This analysis implies that the actual interventions implemented during the COVID-19 crisis were effective but could have generated larger welfare gains in the model if implemented more aggressively using FX reserves.

3.5 Design of sterilized asset purchase programs

The analysis given in the previous sections focuses on a particular type of sterilized asset purchase programs: corporate bond purchases sterilized with FX reserves. Considering this type as the benchmark, this section provides two alternative policy designs. These alternatives deviate from the benchmark along two dimensions: asset purchase and sterilization. Suppose that the economy is in the steady state where banks are constrained (i.e., $\mu > 0$). Among the three different policy designs, which one is the most effective in breaking down the balance sheet effect by relaxing banks' constraints?

[32] The estimated $\hat{\mu}^{\text{data}}$ is 1.93% points per annum. Given the size of shock $\xi_{t=1}$ that matches this estimate, the optimal policy follows the same policy rule but sets $\phi_\mu = 1,500$, while $\phi_\mu = 250$ in the baseline case.

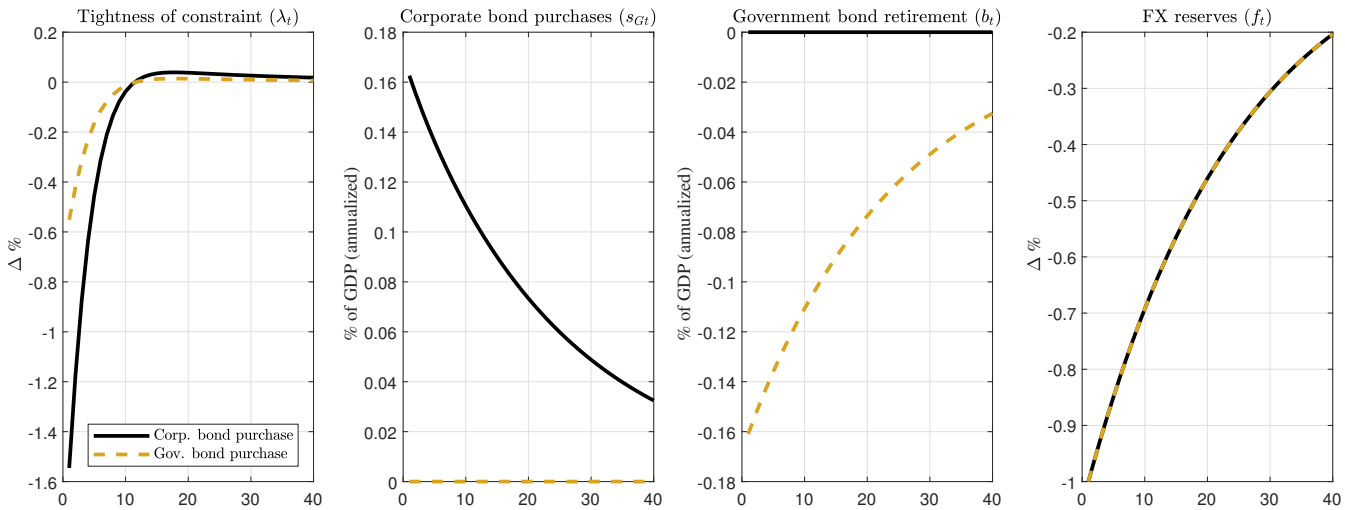

Figure 13: Asset purchase of corporate bond vs. government bond

Note: The figure describes the impulse responses to the discretionary asset purchase by selling 1% of FX reserves from the steady state where banks are constrained. The black solid lines are the responses under corporate bond purchase. The yellow dotted lines are the responses under government bond purchase. The horizontal axis indicates time periods in quarters.

Asset purchase: Corporate vs. Government bond Suppose that the government sells FX reserves following an AR(1) process

$$\frac{f_t}{f} = (1 - \rho_f) + \rho_f \frac{f_{t-1}}{f} - \varepsilon_{ft}$$

where $\rho_f = 0.96$ and the size of ε_{ft} is 0.01 (i.e., 1 percent of FX reserves sale). Using the FX reserves sold, the government can either purchase corporate bonds as in the benchmark

$$q_{kt} \underbrace{s_{Gt}}_{\uparrow} + e_t \underbrace{f_t}_{\downarrow} = q_{bt} \underbrace{b_t}_{\text{constant}}$$

or purchase government bonds (i.e., bond retirement)

$$q_{kt} \underbrace{s_{Gt}}_{\text{constant}} + e_t \underbrace{f_t}_{\downarrow} = q_{bt} \underbrace{b_t}_{\downarrow}$$

Figure 13 compares the effects of corporate bond purchases (black solid lines) and government bond purchases (yellow dotted lines). The government sells 1 percent of FX reserves (fourth panel) for the purchases of both bonds by 0.16 percent of GDP (second and third panels). Notice that when it purchases corporate bonds, the government relaxes banks' leverage constraint 3 times more than it does with government bond

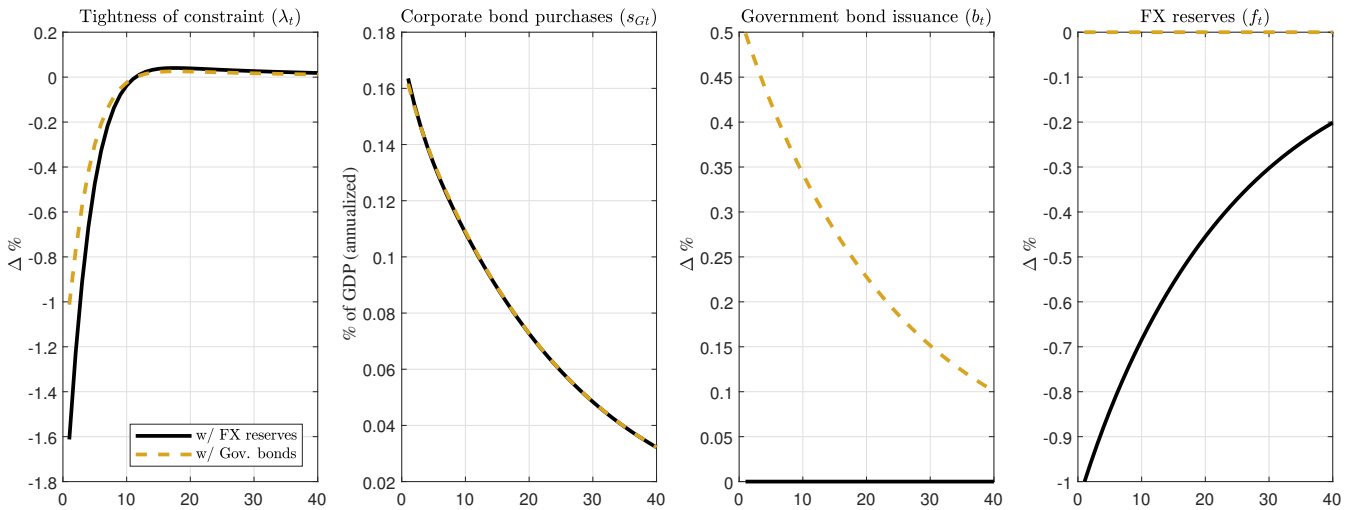

Figure 14: FX reserves vs. other sterilization tools

Note: The figure describes the impulse responses to a discretionary corporate bond purchase from the steady state where banks are constrained. The size of the shock is 0.01. The black solid lines are the responses under sterilization with FX reserves. The yellow dotted lines are the responses under sterilization with other securities. The horizontal axis indicates time periods in quarters.

purchases (first panel). To see why, recall that when the leverage constraint is binding, it can be written as

$$\underbrace{q_{kt}(s_t - s_{Gt}) + \Delta q_{bt} b_t}_{\downarrow} = \bar{l}_t n_t$$

after imposing market clearing conditions for the financial markets. The government asset purchases decrease the left-hand side, relaxing the constraint. Notice that the marginal impact of government bond purchases is only Δ of the marginal impact of corporate bond purchases. Since the corporate bond is considered riskier, a reduction of corporate bond holdings in banks' balance sheet has a greater effect in relaxing the constraint.[33]

Sterilization: Selling FX reserves vs. Issuing government bonds Consider an alternative way of sterilizing asset purchases. Instead of adjusting the asset side of the balance sheet by selling FX reserves, the government can adjust the liability side for sterilization. In particular, suppose that the government issues bonds to sterilize corporate bond purchases. This operation increases both assets and liabilities, expanding the size of the government balance sheet

$$q_{kt} \underbrace{s_{Gt}}_{\uparrow} + e_t \underbrace{f_t}_{\text{constant}} = q_{bt} \underbrace{b_t}_{\uparrow}$$

[33]See also Appendix E.4 that shows that more FX reserves are required if asset purchases are conducted towards government bonds rather than corporate bonds in response to the same risk premium shocks.

Suppose that the government purchases corporate bonds following an AR(1) process

$$\frac{s_{Gt}}{s_G} = (1 - \rho_G) + \rho_G \frac{s_{Gt-1}}{s_G} + \varepsilon_{Gt}$$

where $\rho_G = 0.96$ and the size of ε_{Gt} is 0.01.

Figure 14 plots the effects of corporate bond purchases sterilized by selling FX reserves (black solid lines) and by issuing government bonds (yellow dotted lines). Although the amount of asset purchases is the same in both cases (second panel), the effect on relaxing banks' leverage constraint is about 1.5 times greater when sterilized with FX reserves (first panel). As discussed, the effectiveness of asset purchases depends on the extent to which the leverage constraint is relaxed. When the government issues bonds for sterilization, it is not able to support exchange rates directly. Furthermore, the effect of constraint relaxation is partially offset due to additional government bond issuance. This is because banks must absorb these bonds on their balance sheets, which makes the leverage constraint even more binding. Therefore, selling FX reserves is a more effective way to sterilize asset purchases.

4 Conclusion

Sterilized asset purchase programs are unconventional monetary policies adopted by EMDEs during the recent COVID-19 crisis. The paper develops a theoretical framework to study this new tool by incorporating three key financial frictions: liability dollarization, financial market imperfection, and fear of losing reserves. Sterilized asset purchases mitigate the impact of sudden stops, bringing welfare gains to the economy. In particular, the policy breaks down the balance sheet effect that causes disruptions in domestic banks' funding intermediation.

The main results of this paper provide useful policy guidelines for EMDEs. It is advisable to conduct sterilized asset purchase programs in a sudden stop when domestic banks are constrained. The most effective way of implementing the policy is to purchase corporate bonds and use FX reserves for sterilization. It is important to acknowledge that there are policy trade-offs regarding the degree of intervention. Purchasing assets on a large scale can prevent deep contractions in real activity. However, it may also slow down the recovery process of the economy. Finally, hoarding large enough FX reserves expands the policy room for asset purchase programs during sudden stops.

For future research, it would be interesting to extend the model to include nominal rigidities. This extension would provide insights on how to implement policies given the comprehensive set of conventional and unconventional monetary policy tools.

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Appendix

A Bank's problem

First note that, combining (2) and (4), the law of motion for bank's net worth is

$$n_{jt+1} = \left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) q_{kt} s_{Bjt} + \left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) q_{bt} b_{Bjt} + R_{t+1}\frac{e_{t+1}}{e_t} n_{jt} \tag{19}$$

Since the problem is linear, let $V_t \equiv v_t n_{jt}$, a linear function of n_{jt} with coefficient v_t to be determined. Using (19), the banks' problem can be rewritten as

$$v_t = \max_{s_{Bjt}, b_{Bjt}} \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left[\left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) \frac{q_{kt} s_{Bjt}}{n_{jt}} + \left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) \frac{q_{bt} b_{Bjt}}{n_{jt}} + R_{t+1}\frac{e_{t+1}}{e_t} \right]$$

s.t.

$$v_t \geq \theta \left(\frac{q_{kt} s_{Bjt}}{n_{jt}} + \Delta \frac{q_{bt} b_{Bjt}}{n_{jt}} \right) \tag{20}$$

where $\Omega_{t+1} = (1 - \chi) + \chi v_{t+1}$.

Let v_t denote the Lagrange multiplier attached to (20) and define $\lambda_t \equiv v_t/(1 + v_t)$. The first-order conditions are

$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \theta \lambda_t$$
$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \Delta \theta \lambda_t$$

and the complementary slackness conditions are $\lambda_t \geq 0$, $v_t \geq \theta l_{jt}$, and

$$\lambda_t \left(v_t - \theta l_{jt}\right) = 0$$

where

$$l_{jt} \equiv \frac{q_{kt} s_{Bjt} + \Delta q_{bt} b_{Bjt}}{n_{jt}}$$

is the leverage ratio. Combining the first-order conditions, one can find the no-arbitrage condition as

$$\Delta \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{kt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right) = \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{bt+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right)$$

By plugging this equation back in the objective function,

$$v_t = v_{kt} l_{jt} + v_{nt}$$

where $\nu_{kt} = \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} (R_{kt+1} - R_{t+1} e_{t+1}/e_t)$ and $\nu_{nt} = \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} R_{t+1} e_{t+1}/e_t$.

Consider the case where (20) is binding. In this case, $\lambda_t > 0$ and

$$\theta l_{jt} = \nu_{kt} l_{jt} + \nu_{nt}$$

Hence, the leverage ratio attains the maximum value

$$\begin{aligned} l_{jt} &= \frac{\nu_{nt}}{\theta - \nu_{kt}} \\ &\equiv \bar{l}_t \end{aligned}$$

the coefficient for the value function becomes

$$\nu_t = \theta l_{jt}$$

and $\lambda_t = \nu_{kt}/\theta$, or

$$\lambda_t = 1 - \frac{\nu_{nt}}{\theta l_{jt}}$$

using the leverage ratio.

Consider now the case where (20) is not binding. In this case, $\lambda_t = 0$ implying that $\nu_{kt} = 0$. Hence, the coefficient for the value function becomes

$$\nu_t = \nu_{nt}$$

Hence, combining the two cases,

$$\nu_t = \frac{\nu_{nt}}{1 - \lambda_t}$$

where

$$\lambda_t = \max \left\{ 0, 1 - \frac{\nu_{nt}}{\theta l_{jt}} \right\}$$

Assuming symmetric leverage ratios across banks in equilibrium, $l_{jt} = l_t$ for all j, the aggregate leverage ratio is

$$l_t = \frac{q_{kt} s_{Bt} + \Delta q_{bt} b_{Bt}}{n_t}$$

where $n_t \equiv \int_0^\zeta n_{jt} dj$, $s_{Bt} \equiv \int_0^\zeta s_{Bjt} dj$, $b_{Bt} \equiv \int_0^\zeta b_{Bjt} dj$ are the aggregate net worth, corporate bond holdings, and government bond holdings. Notice that when (20) is binding, $l_t = \bar{l}_t$, implying that the aggregate demand for assets is tied to the value of net worth,

$$q_{kt} s_{Bt} + \Delta q_{bt} b_{Bt} = n_t \bar{l}_t$$

Aggregating over (19), the aggregate net worth for existing banks in period t is

$$n_{ot} = \left(R_{kt} - R_t \frac{e_t}{e_{t-1}}\right) q_{kt-1} s_{Bt-1} + \left(R_{bt} - R_t \frac{e_t}{e_{t-1}}\right) q_{bt-1} b_{Bt-1} + R_t \frac{e_t}{e_{t-1}} n_{t-1}$$

B Appendix: Balance of payments derivation

To derive the balance of payments, first combine the budget constraints for households (1) and the aggregate banks (3). Given the fact that in equilibrium (i) $s_t = k_t$ and $s_{Bt} = k_t - s_{Gt}$; (ii) the profits from home good producers and capital storage firms are zero,

$$\begin{aligned} & c_t + q_{kt}(k_t - s_{Gt}) + e_t R_t d_{Ft-1} + q_{bt} b_{Bt} \\ & \quad = w_t h_t + R_{kt} q_{kt-1}(k_{t-1} - s_{Gt-1}) + R_{bt} q_{bt-1} b_{Bt-1} + e_t d_{Ft} + \Pi_t^c + T_t \end{aligned}$$

where Π_t^c is the profits from capital producers. Since $\Pi_t^c = q_{kt}(k_t - (1-\delta)k_{t-1}) - i_t$ and $p_{Ht} y_{Ht} = z_t k_{t-1} + w_t h_t$, the above equation is reduced to

$$c_t + i_t - q_{kt} s_{Gt} + e_t R_t d_{Ft-1} + q_{bt} b_{Bt} = p_{Ht} y_{Ht} - R_{kt} q_{kt-1} s_{Gt-1} + R_{bt} q_{bt-1} b_{Bt-1} + e_t d_{Ft} + T_t$$

Combining it with the government budget constraint (13),

$$p_{Ht} g_t + c_t + i_t + e_t(R_t d_{Ft-1} - R_t^* f_{t-1}) = p_{Ht} y_{Ht} + e_t(d_{Ft} - f_t)$$

Since $c_t = p_{Ht} c_{Ht} + e_t c_{Ft}$ and $i_t = p_{Ht} i_{Ht} + e_t i_{Ft}$, the above equation becomes

$$e_t(R_t d_{Ft-1} - R_t^* f_{t-1}) = p_{Ht} x_t - e_t m_t + e_t(d_{Ft} - f_t)$$

after imposing the market clearing condition for home goods (15).

Hence, using the definition of trade balance (16), one can derive the balance of payments

$$tb_t = -e_t(d_{Ft} - f_t) + e_t(R_t d_{Ft-1} - R_t^* f_{t-1})$$

and the national income identity

$$y_t = p_{Ht} g_t + c_t + i_t + tb_t$$

where $y_t \equiv p_{Ht} y_{Ht}$.

C Equilibrium conditions

$$mu_t = \left(c_t - \frac{\kappa_h\, h_t^{1+\varphi}}{1+\varphi}\right)^{(-\sigma)}$$

$$\Lambda_{t,t+1} = \frac{\beta\, mu_{t+1}}{mu_t}$$

$$1 = \mathbb{E}_t \Lambda_{t,t+1}\, R_{t+1}\, \frac{e_{t+1}}{e_t}$$

$$\kappa_h\, h_t^{\varphi} = w_t$$

$$c_{Ht} = (1-\gamma)(p_{Ht})^{-\eta} c_t$$

$$c_{Ft} = \gamma (e_t)^{-\eta} c_t$$

$$y_{Ht} = a_t\, k_{t-1}^{\alpha}\, h_t^{1-\alpha}$$

$$(1-\alpha)\, p_{Ht} y_{Ht} = w_t h_t$$

$$\alpha p_{Ht} y_{Ht} = z_t k_{t-1}$$

$$k_t = (1-\delta)\, k_{t-1} + i_t \left(1 - \frac{\kappa_i}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2\right)$$

$$1 = q_{k_t}\left(1 - \frac{\kappa_i}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2 - \kappa_i\left(\frac{i_t}{i_{t-1}} - 1\right)\frac{i_t}{i_{t-1}}\right) + \kappa_i \mathbb{E}_t \Lambda_{t,t+1}\, q_{k_{t+1}}\left(\frac{i_{t+1}}{i_t} - 1\right)\left(\frac{i_{t+1}}{i_t}\right)^2$$

$$i_{Ht} = (1-\gamma)(p_{Ht})^{-\eta} i_t$$

$$i_{Ft} = \gamma (e_t)^{-\eta} i_t$$

$$R_{k_t} = \frac{z_t + (1-\delta)\, q_{k_t}}{q_{k_{t-1}}}$$

$$R_{b_t} = \frac{\Xi + \varrho\, q_{b_t}}{q_{b_{t-1}}}$$

$$l_t = \frac{q_{k_t}\, s_{B_t} + \Delta\, q_{b_t} b_{B_t}}{n_t}$$

$$q_{k_t}\, s_{Bt} + q_{b_t}\, b_{Bt} = n_t + e_t\, d_t$$

$$n_t = \chi \left(\left(R_{k_t} - R_t \frac{e_t}{e_{t-1}} \right) q_{k_{t-1}}\, s_{Bt-1} + \left(R_{b_t} - R_t \frac{e_t}{e_{t-1}} \right) q_{b_{t-1}}\, b_{Bt-1} + R_t \frac{e_t}{e_{t-1}}\, n_{t-1} \right) + \iota \left(q_{k_{t-1}}\, s_{Bt-1} + \Delta q_{b_{t-1}} b_{Bt-1} \right)$$

$$\Omega_t = 1 - \chi + \chi \nu_t$$

$$\nu_t = \frac{\nu_{nt}}{1 - \lambda_t}$$

$$\lambda_t = \max \left\{ 0, 1 - \frac{\nu_{nt}}{\theta l_t} \right\}$$

$$\nu_{nt} = \mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} R_{t+1} e_{t+1} / e_t$$

$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{kt+1} - R_{t+1} \frac{e_{t+1}}{e_t} \right) = \theta \lambda_t$$

$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{bt+1} - R_{t+1} \frac{e_{t+1}}{e_t} \right) = \Delta \theta \lambda_t$$

$$y_{Ht} = g_t + (1 - \gamma)\, {p_{Ht}}^{(-\eta)}\, (c_t + i_t) + \gamma^*\, tot_t^{\eta} y_t^*$$

$$y_t = p_{Ht} y_{Ht}$$

$$y_t = c_t + i_t + p_{Ht} g_t - e_t\, (d_{F_t} - f_t) + e_t\, \left(R_t\, d_{F_{t-1}} - R_t^*\, f_{t-1} \right)$$

$$1 = (1 - \gamma)\, {p_{Ht}}^{1-\eta} + \gamma\, e_t^{1-\eta}$$

$$tot_t = \frac{e_t}{p_{Ht}}$$

$$xp_t = p_{Ht}\, \gamma^*\, tot_t^{\eta}\, y_t^*$$

$$mp_t = \gamma\, e_t^{1-\eta}\, (c_t + i_t)$$

$$tb_t = xp_t - mp_t$$

$$ca_t = tb_t + e_t\, \left(f_{t-1}\, \left(R_t^* - 1 \right) - d_{F_{t-1}}\, (R_t - 1) \right)$$

$$k_t = s_t$$

$$s_t = s_{Bt} + s_{Gt}$$

$$b_t = b_{Bt}$$

$$d_{Ht} = d_t - d_{Ft}$$

$$q_{kt}\, s_{Gt} + e_t f_t = q_{bt} b_t$$

$$f_t = (1 - \Gamma_t) f$$

$$b_t = b$$

$$g_t = g$$

$$\Gamma_t = \min\left\{\phi_\mu(\mu_t - \mu), \bar{\Gamma}\right\}$$

$$\mu_t = \mathbb{E}_t\left(R_{k\,t+1} - R_{t+1}\frac{e_{t+1}}{e_t}\right)$$

$$a_t = a$$

$$R_t = R_t^* + \psi\left(\exp\left(\frac{e_t\ (d_{Ft} - f_t)}{y_t} - \frac{e(d_F - f)}{y}\right) - 1\right) + \xi_t$$

$$\xi_t = \rho_\xi\, \xi_{t-1} + \varepsilon_{\xi t}$$

$$y_t^* = y^*$$

$$R_t^* = R^*$$

D Data appendix

The data for five countries (Chile, Colombia, Korea, Philippines, Turkey) are from the International Financial Statistics (IFS) and Financial Soundness Indicator (FSI) at the International Monetary Fund, the World Development Index (WDI) and Global Economic Monitor (GEM) at the World Bank, Main Economic Indicators (MEI) at the Organization for Economic Co-operation and Development, and the central banks. The data series from each source are listed below.

- IFS
  - Gross Domestic Product, Real, Seasonally Adjusted, Domestic Currency, Millions (NGDP_R_SA_XDC)
  - Gross Domestic Product, Real, Domestic Currency, Millions (NGDP_R_XDC)
  - Private Sector Final Consumption Expenditure, Real, Seasonally Adjusted, Domestic Currency, Millions (NCP_R_SA_XDC)
  - Private Sector Final Consumption Expenditure, Real, Domestic Currency, Millions (NCP_R_XDC)
  - Gross Capital Formation, Real, Seasonally Adjusted, Domestic Currency, Millions(NI_R_SA_XDC)[34]
  - Gross Capital Formation, Real, Domestic Currency, Millions (NI_R_XDC)
  - Exports of Goods and Services, Real, Seasonally Adjusted, Domestic Currency, Millions (NX_R_SA_XDC)[35]
  - Exports of Goods and Services, Real, Domestic Currency, Millions (NX_R_XDC)
  - Imports of Goods and Services, Real, Seasonally Adjusted, Domestic Currency, Millions (NM_R_SA_XDC)[36]
  - Imports of Goods and Services, Real, Domestic Currency, Millions (NM_R_XDC)
  - Balance of Payments, Supplementary Items, Current Account, Net (excluding exceptional financing), US Dollars, Millions (BCAXF_BP6_USD)
  - Exchange Rates, Real Effective Exchange Rate based on Consumer Price Index, Index (EREER_IX)[37]
  - Exchange Rates, Domestic Currency per U.S. Dollar, Period Average, Rate (ENDA_XDC_USD_RATE)
  - Prices, Consumer Price Index, All items, Index (PCPI_IX)
  - Financial, Interest Rates, Lending Rate, Percent per annum (FILR_PA)[38]
  - Financial, Interest Rates, Deposit, Percent per annum (FIDR_PA)
- FSI
  - Deposit Takers, Earnings and Profitability, Capital (FSDERE_XDC)
  - Deposit Takers, Earnings and Profitability, Total Assets (FSDERA_XDC)

[34] This series is not available for Turkey and Philippines.
[35] This series is not available for Philippines.
[36] This series is not available for Philippines.
[37] This series is not available for Korea and Turkey. For these countries, I construct real exchange rates with nominal exchange rates and the consumer price indices.
[38] This series is not available for Turkey.

  - Deposit-takers, Liabilities, Debt, Other liabilities (FS_ODX_LDO_XDC)

- WDI
  - General government final consumption expenditure, % of GDP (NE.CON.GOVT.ZS)
  - Exports of goods and services, % of GDP (NE.EXP.GNFS.ZS)
  - External debt stocks, total, DOD, current US$ (DT.DOD.DECT.CD)[39]
  - Total reserves minus gold, current US$ (FI.RES.XGLD.CD)
  - GDP, current US$ (NY.GDP.MKTP.CD)
  - Balance of Payments, Supplementary Items, Current Account, Net (excluding exceptional financing), US Dollars, Millions (BN.CAB.XOKA.GD.ZS)

- GEM
  - J.P. Morgan Emerging Markets Bond Spread, EMBI+ (EMBIG)

- MEI
  - Long-term government bond yields, 10-year (IRLTLT01)[40]

- Bank of Chile
  - External debt to market value (millions of dollars)

- Bank of Korea
  - 1.3.2.2. Market Interest Rates, Yields of Treasury Bonds(3-year), Percent Per Annum
  - 1.3.2.2. Market Interest Rates, Yields of Corporate Bonds : O.T.C (3-year, AA-), Percent Per Annum
  - 2.6.2.1. External Debt, 1. Short-term, Mil.U$
  - 2.6.2.1. External Debt, 2. Long-term, Mil.U$

[39]Since this series is not available for Chile and Korea, I use data directly from the central banks for the two countries.

[40]The series includes data for Chile, Colombia, and Korea. The data for Turkey is not available. Philippines is not part of OECD.

E Robustness checks

E.1 Balance sheet effect

To see in more detail the amplification of the sudden stop through the balance sheet effect, I compare impulse responses with and without banks' leverage constraint. Figure E.1 describes two sets of impulse responses without policy in a sudden stop. The blue solid lines are the same responses as in Figures 3 and 4 from Section 3.2. In addition to these responses, the figure also depicts the ones without the leverage constraint imposed on the banks (red dotted lines). The comparison between the two highlights the role of banks' leverage constraint implementing the balance sheet effect in a sudden stop.

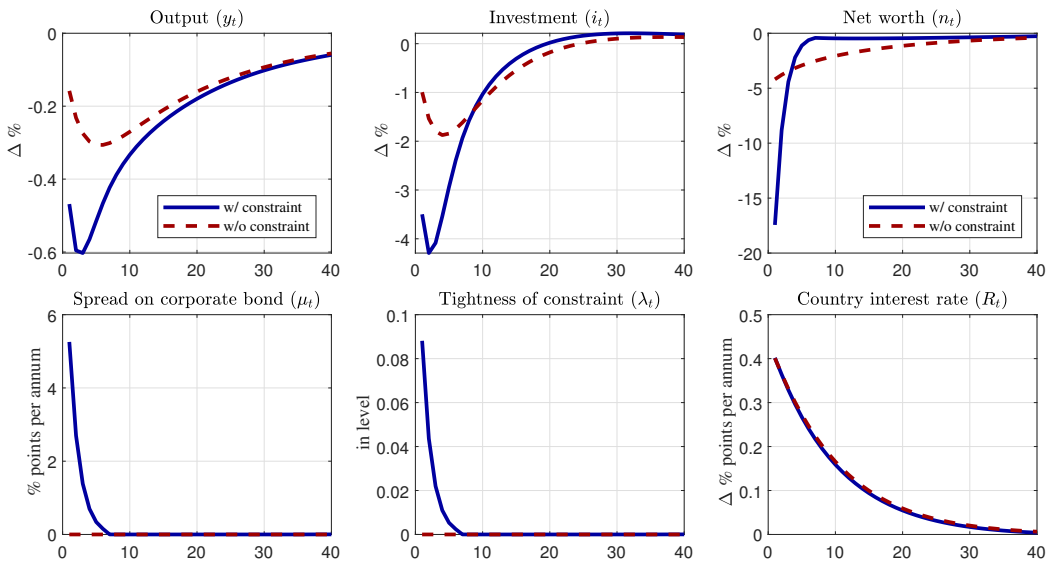

Figure E.1: Impulse responses in a sudden stop: with and without leverage constraint

Note: The figure describes impulse responses to 0.1 percentage point of risk premium shock from the steady state. Compared to the benchmark responses in blue solid lines where banks' leverage constraint is imposed, the red solid line responses without leverage constraint imposed display milder contractions. In both cases, the policy is not implemented. The horizontal axis indicates time periods in quarters.

Specifically, the figure shows that, for the same size of risk premium shock, the economy experiences much milder impact of the sudden stop, in the absence of balance sheet effect. Without the amplification mechanism, banks lose only about 5 percent of net worth (top-right panel). Since banks are not constrained (bottom-middle panel), they can absorb any excess returns on the corporate bond (bottom-left panel). As a result, the contractions in output and investment are much milder. This counterfactual analysis demonstrates that the balance sheet effect is the key mechanism that amplifies the adverse effect of the sudden stop.

E.2 Partial currency mismatch

Consider the case where currency mismatch in banks' balance sheet is partial. In particular, assume that the banks borrow from households only in domestic currency. Accordingly, the Euler equation derived

from the household problem becomes

$$1 = \mathbb{E}_t \Lambda_{t,t+1} R_{dt+1}$$

where R_{dt} is the return on domestic currency deposits. Bank j's balance sheet becomes

$$q_{kt} s_{Bjt} + q_{bt} b_{Bjt} = n_{jt} + d_{jHt} + e_t d_{jFt} \tag{21}$$

where the mismatch of currency denomination is partial, i.e., not all liabilities are in foreign currency. Bank j's budget constraint is

$$e_t R_t d_{jt-1} + q_{kt} s_{Bjt} + q_{bt} b_{Bjt} = d_{jHt} + e_t d_{jFt} + R_{kt} q_{kt-1} s_{Bjt-1} + R_{bt} q_{bt-1} b_{Bjt-1} \tag{22}$$

Combining (21) and (22),

$$n_{jt} = R_{kt} q_{kt-1} s_{Bjt-1} + R_{bt} q_{bt-1} b_{Bjt-1} - R_{dt} d_{jHt-1} - e_t R_t d_{jFt-1} \tag{23}$$

Importantly, assume the leverage constraint as

$$V_t(n_{jt}) \geq \theta(q_{kt} s_{Bjt} + \Delta q_{bt} b_{Bjt} - \Delta_H d_{jHt}) \tag{24}$$

where $\Delta_H > 0$ meaning that it is harder for banks to abscond with domestic currency deposit relative to foreign currency deposit.[41] The first-order conditions become

$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{kt+1} - R_{t+1} \frac{e_{t+1}}{e_t} \right) = \theta \lambda_t$$
$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{bt+1} - R_{t+1} \frac{e_{t+1}}{e_t} \right) = \Delta \theta \lambda_t$$
$$\mathbb{E}_t \Lambda_{t,t+1} \Omega_{t+1} \left(R_{dt+1} - R_{t+1} \frac{e_{t+1}}{e_t} \right) = \Delta_H \theta \lambda_t$$

When the constraint is binding (i.e., $\lambda > 0$), the last equation indicates excess costs of borrowing in domestic currency over foreign currency, yielding the violation of uncovered interest parity (UIP) condition. The leverage ratio is defined as below.

$$l_{jt} \equiv \frac{q_{kt} s_{Bjt} + \Delta q_{bt} b_{Bjt} - \Delta_H d_{jHt}}{n_{jt}} \tag{25}$$

Figure E.2 shows in partial dollarization the sudden stop brings a milder impact, compared to full dollarization.[42] This is because, under partial dollarization, the balance sheet effect is milder, and hence, banks are constrained not as much as in the full dollarization case.

[41] See also similar assumptions made by Kitano and Takaku (2020) and Mimir and Sunel (2019, 2021)

[42] The newly introduced parameter Δ_H is set to 0.1 in the simulation, which is within the range commonly used in the literature. For example, Mimir and Sunel (2021) uses a value of 0.1769 for 13 EMDEs.

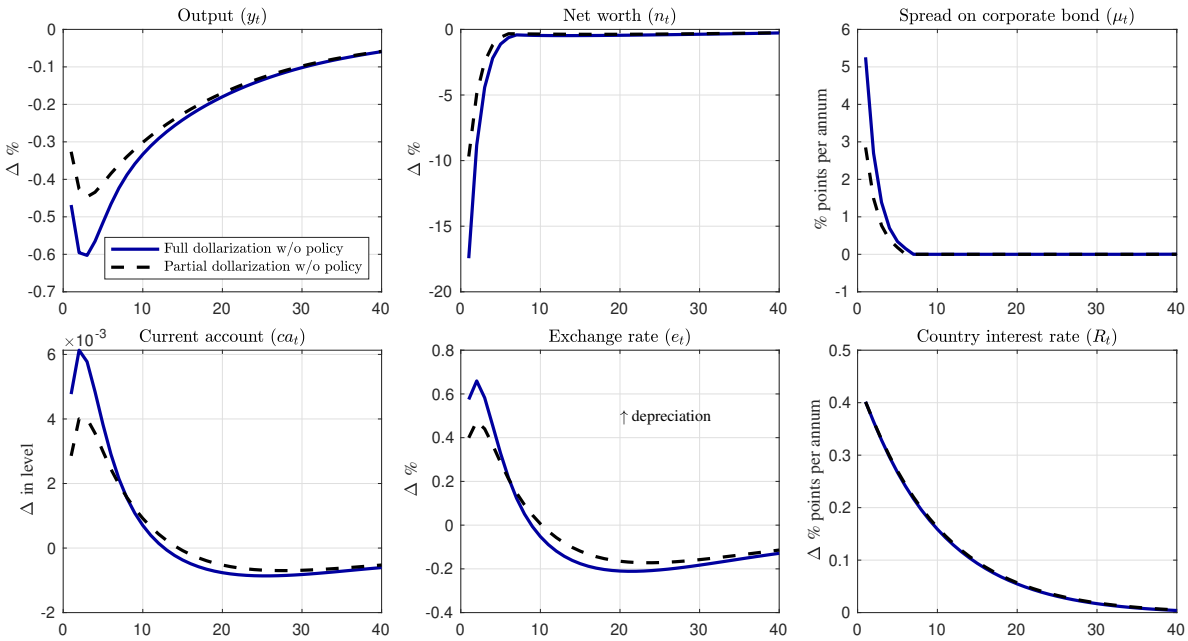

Figure E.2: Full vs. Partial dollarization

Note: The figure describes the impulse responses to the 0.1 percentage point of the risk premium shock. It compares the full dollarization (blue straight line) with the partial dollarization (black dotted line).

E.3 On slow recovery

Figure E.3 describes the marginal changes in FX reserve sales, corporate bond spreads, and net worth growth to the additional asset purchases as a percentage of GDP. It is measured at the peak. As the government intervenes to purchase more assets, it uses more of FX reserves for sterilization, which leads to a larger decrease in corporate bond spreads and a smaller net worth growth in the recovery phase.

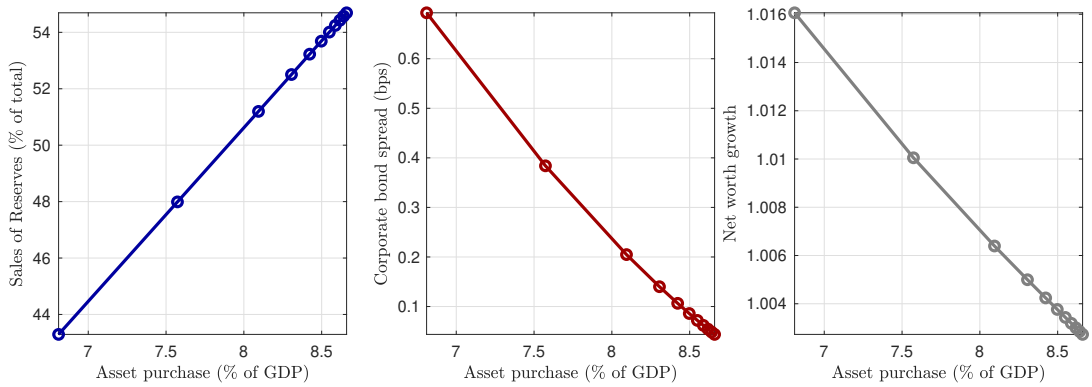

Figure E.3: Marginal changes in FX reserves, Corporate bond spread, and Net worth growth

Note: The figure describes the marginal changes in FX reserve sales, corporate bond spreads, and net worth growth to the additional asset purchases as a percentage of GDP in response to the 0.1 percentage point of the risk premium shock. It is measured at the peak.

E.4 The need for FX reserves under different asset purchases

Recall that Figure 13 in Section 3.5 shows that corporate bond purchases are more cost-effective than government bond purchases. For the same amount of FX reserves, purchasing corporate bonds is more effective in relaxing banks' leverage constraints. This section provides a similar analysis from a different angle by examining if the need for FX reserves depends on which asset is purchased.

Figure E.4 describes the impulse responses where the government purchases corporate bonds (in green straight lines) and government bonds (in orange dotted lines) in response to the same risk premium shock. It shows that the government uses more FX reserves for sterilization when it purchases government bonds than when it purchases corporate bonds. This is because purchasing government bonds is not as effective in relaxing banks' leverage constraints as much as purchasing corporate bonds. As a result, the corporate bond spread (a measure for financial market stress) remains high, leading to larger amounts of asset purchases following the policy rule in equation (14). Despite more asset purchases and accompanied FX reserve sales, the government bond purchases are less effective, and hence, banks' net worth deteriorates more when the risk premium shock hits.

The key takeaway from this exercise is that the need for FX reserves depends on which asset is purchased. The policy rule depends on the corporate bond spread which in turn is endogenously influenced by asset purchases. Since purchasing corporate bonds is a more effective in reducing spreads, the government ends up using smaller amount of FX reserves when it chooses to purchase corporate bonds rather than government bonds.

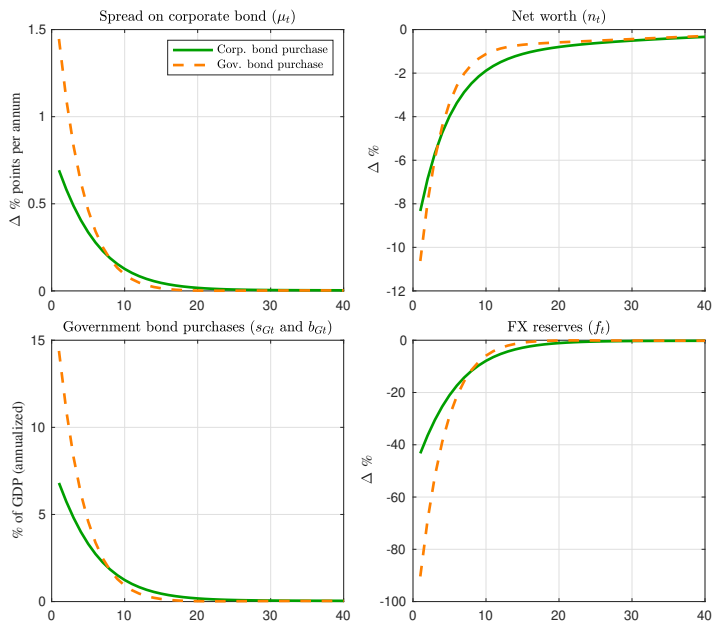

Figure E.4: Use of FX reserves under different asset purchases

Note: The figure describes the impulse responses for different types of asset purchases: corporate bond (in green straight lines) and government bond (orange dotted lines) in response to the 0.1 percentage point of the risk premium shock from the steady state where banks are not constrained. In both cases, the government is following the same policy rule in equation (14) with ϕ_μ set to 250. The horizontal axis indicates time periods in quarters.